

E-commerce & marketplaces

JAIPUR, THE FIRST NIGHT OF THE OCTOBER SALE, 11:58 PM

Pooja, 26, has had a kurta in her cart for nine days. At midnight its price drops from ₹1,899 to ₹1,299. She isn't sure whether she is a medium or a large, so she orders both. She will keep one and send the other back. Across India, shoppers are doing the same thing in the first minutes of the sale. The app shows a sale and a spike in sales. The finance team will see the cost three weeks later, when a third of the fashion orders come back.

Try it: you are the PM for kurtas on a fashion app, and 30% of them come back. Is that a problem worth solving? For whom, and how would you know? Think for five minutes before reading on.

How a ₹1,299 kurta makes (a little) money

Pooja pays ₹1,299. The kurta belongs to a seller in Jaipur's textile market. It sits in the app's warehouse, a delivery partner carries it 6 km, and the app earns fees on the way. If Pooja keeps it, the app is left with about ₹195. If she returns it, the fees are refunded and about ₹250 of trucking, checking and discounting is lost. E-commerce is a business where the order you *keep* is the only one that counts.

~\$65 bn India's online retail sales (GMV) in 2025, up about 20% on 2024	290–300 mn Indians who shopped online in 2025, second only to China	264 mn people who bought on Meesho at least once in FY26, up 33%	25–40% share of online fashion orders returned in India (industry estimates, 2025)
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Who does what, and who pays whom

An online shop in India is a **marketplace**: an app where independent sellers list goods and the app earns fees for bringing them buyers. Five parties touch every order. The **seller** owns the stock and sets the price. The **platform** (Flipkart, Amazon, Myntra, Meesho) runs the app, the search results and often the warehouse. A **logistics partner** carries the parcel (Ekart for Flipkart, Amazon's own fleet, Delhivery or Shadowfax for many others). A **payment rail** moves the money (UPI, cards, EMI or cash on delivery). And **brands** pay the platform for ads so their products show up first. Follow the money in Figure 3.1.

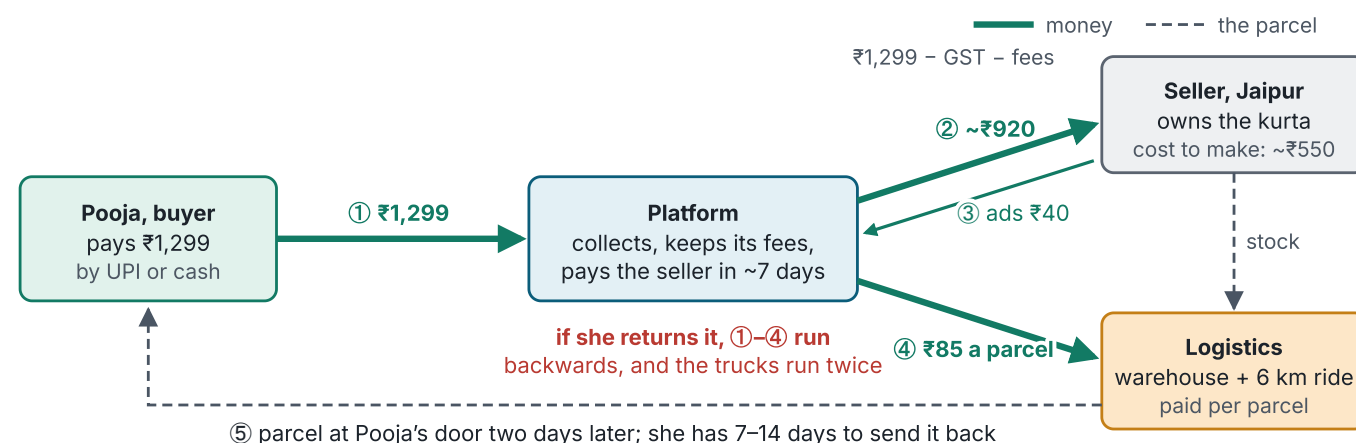


Figure 3.1 · Who pays whom on one marketplace order. The platform sits in the middle of every rupee: it collects from the buyer, keeps its fees and the seller's ad spend, pays the delivery partner, and passes the rest to the seller a week later. *Illustrative fashion order; fee levels typical of large Indian marketplaces, 2025–26. GST simplified.*

Why India's big platforms must be marketplaces. Indian rules on foreign investment (FDI) allow a foreign-owned company to run an e-commerce *marketplace* but not to own the goods it sells online. Amazon is American-owned and Flipkart has been owned by Walmart since 2018, so both must work through independent sellers. **Press Note 3 (2016)** set this rule, and **Press Note 2 (2018)** tightened it: no exclusive deals, and no controlling the stock of sellers the platform has a stake in. That is why Amazon had to exit its stakes in large sellers such as Cloudtail. Indian-owned companies (Reliance's JioMart, Tata's BigBasket, and Blinkit since 2025) may own stock.

The unit economics, worked in rupees

Take Pooja's kurta from the platform's side. The seller pays a commission of about 20% on the price before GST (≈ ₹250), a shipping and fixed fee (≈ ₹90) and, this month, ≈ ₹40 of ads. So the platform takes in about ₹380. Out of that it pays the delivery partner ≈ ₹85, funds a ₹60 sale discount, pays ≈ ₹15 for payments and cash handling, and spends ≈

₹25 per order on support and technology. About ₹195 is left (Figure 3.2). That ₹195 is the order's **contribution** (money left after the costs this one order caused).

Platform's side of one ₹1,299 kurta, kept by the customer

The same kurta, returned: fees refunded, and the system is out about ₹250

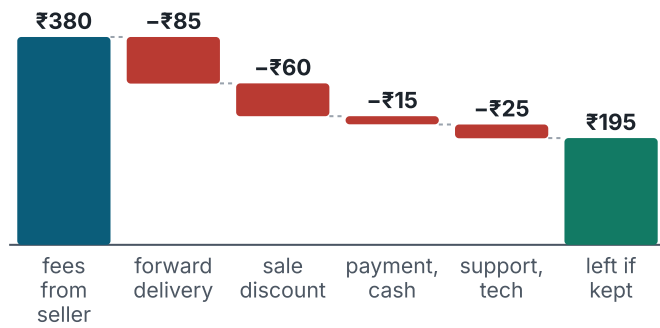


Figure 3.2 · A kept order. The ₹380 of fees shrink to ₹195 once delivery, the sale discount and payments are paid for. *Illustrative; fee rates typical of 2025–26.*

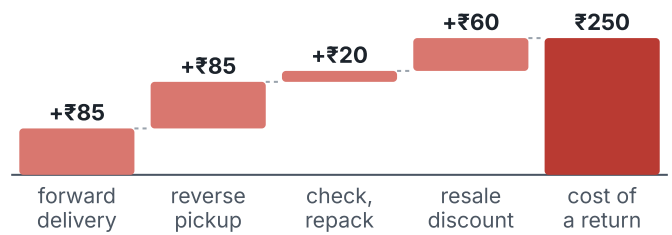


Figure 3.3 · A returned order. The fees go back to the seller, but two trips, a quality check and a discounted resale cost about ₹250. *Illustrative; who bears each rupee depends on the platform's fee rules.*

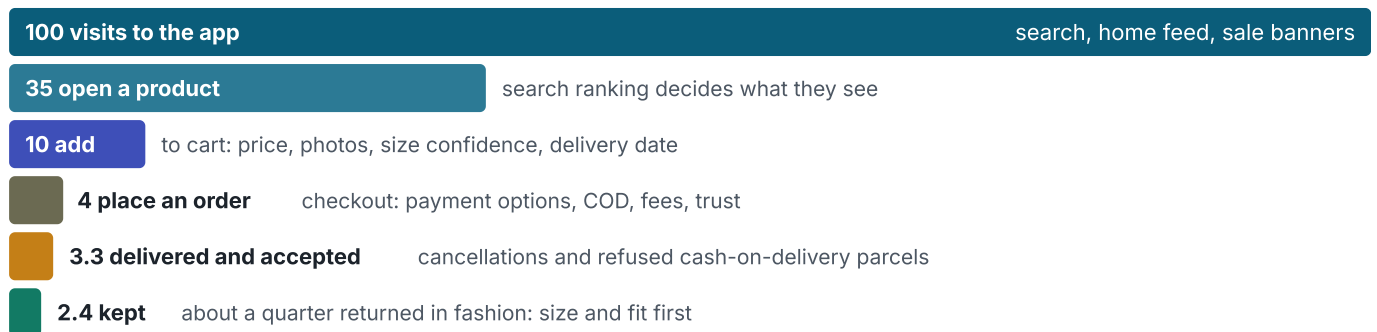
Now add returns. Out of every 10 kurtas at a 30% return rate: 7 kept $\times$ ₹195 = ₹1,365, minus 3 returned $\times$ ₹250 = ₹750. That leaves ₹615, or about ₹62 an order. At a 20% return rate: 8 $\times$ ₹195 – 2 $\times$ ₹250 = ₹1,060, or ₹106 an order. **Ten points off the return rate is worth 70% more money per order.** No price rise or ad campaign comes close. That is why the cold open's question matters, and why fashion apps obsess over size charts.

KEY IDEA · PLACED IS NOT DELIVERED, AND DELIVERED IS NOT KEPT

Meesho placed 2,668 million orders in FY26 but delivered 1,710 million: **36% of orders never reached a customer who accepted them.** The gap is cancellations and **RTO** (return to origin: a parcel refused at the door, often a cash-on-delivery order the buyer no longer wants, sent all the way back). Then some delivered orders are returned. So always ask which number a chart shows: placed, delivered or kept. *Meesho FY26 results, May 2026.*

The funnel: where shoppers drop off

A shopping app is a long funnel (a sequence of steps where some people drop out at each one). The first half is about **finding** the right thing; the second half is about **getting it and keeping it**. Most PM work in e-commerce targets one step of Figure 3.4.



Finding (top three rows) is a search and ranking problem; keeping (bottom two) is a trust, size and logistics problem.

Figure 3.4 · From 100 visits to 2.4 kept orders. Each step loses people for different reasons, so each has a different owner and a different fix. Only the last row pays. *Illustrative rates for a fashion app; the delivered and returned steps are consistent with Meesho FY26 and industry return-rate estimates.*

Metrics that matter

Metric	What it means, in one line	Pooja's kurta, or a typical value
GMV (gross merchandise value)	Value of all orders placed, before cancellations and returns	₹1,299 $\times$ 2 sizes = ₹2,598 counted on sale night
NMV (net merchandise value)	GMV after cancellations and returns: what customers actually kept	₹1,299; half of sale-night GMV
AOV (average order value)	GMV $\div$ number of orders	Meesho ~₹240 (FY26 NMV $\div$ delivered orders); fashion apps roughly ₹1,000–1,500 (estimate)
Take rate	Platform revenue $\div$ GMV: the share of each rupee the platform keeps	₹380 $\div$ ₹1,299 $\approx$ 29% on this order
Conversion rate	Orders $\div$ visits	4 per 100 visits in Figure 3.4

Return rate · RTO rate	Share of delivered orders sent back · share of dispatched orders refused at the door	25–40% in fashion · about 20–25% of COD orders (industry estimates)
Ad revenue % of GMV	How much sellers and brands pay to be seen	Myntra: ads were 15% of FY25 revenue
Repeat rate	Share of buyers who order again within, say, 90 days	The number behind every loyalty programme

The players, and where they are now

Player	Position	Latest numbers	2025–26 move
Flipkart (Walmart)	The biggest by sales; phones, appliances, the October sale; Tier-2 India	FY25 revenue ₹20,493 Cr (+14%), loss ₹1,494 Cr	Zero commission under ₹1,000 (Nov 2025); Minutes at ~1,000 dark stores
Amazon India	Metro households, Prime, fast delivery, trusted electronics	Not reported separately; Amazon's ads earned \$19.8 bn worldwide in Apr–Jun 2026	Zero referral fee under ₹1,000 (Mar 2026); Amazon Now 10-minute delivery
Meesho	Unbranded value goods, first-time buyers, zero commission	FY26 revenue ₹12,626 Cr (+34%), loss ₹1,358 Cr, 264 mn buyers	Listed on the stock exchanges, Dec 2025
Myntra (Flipkart)	Fashion and beauty, brand-led, ages 18–35	FY25 revenue ₹6,043 Cr, profit ₹548 Cr	M-Now: fashion in 30 minutes from city stores
Others	Nykaa (beauty), JioMart, Tata Neu, ONDC (open network)	Quick commerce is taking everyday purchases: Blinkit alone had 2,243 dark stores at the end of FY26	

Sources: company results and filings (Meesho May 2026, Myntra and Flipkart RoC filings via Entrackr, Amazon Q2 2026 release); Bain & Flipkart, *How India Shops Online 2025*.

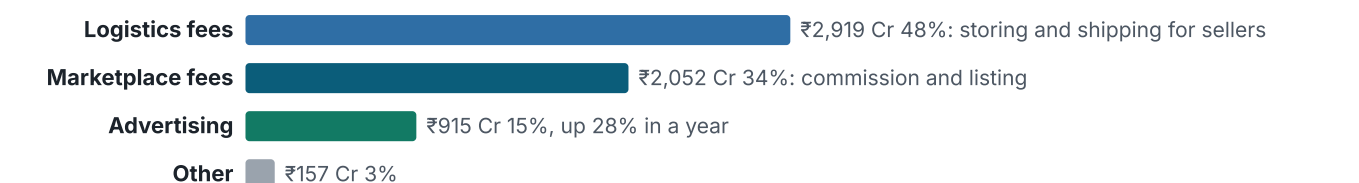


Figure 3.5 · How Myntra earned its ₹6,043 Cr in FY25. Half its revenue is charging sellers for logistics; ads (₹915 Cr, up 28%) grew fastest and cost almost nothing to deliver, which is why a fashion app turned a ₹548 Cr profit. *Myntra Designs FY25 filing via Entrackr, Sep 2025. “Other” derived as the remainder.*

Three turning points

2016–18: the FDI rules. Press Notes 3 and 2 forced foreign-owned platforms to be pure marketplaces. Amazon and Flipkart rebuilt around third-party sellers, and “seller services” (warehousing, loans, ads for sellers) became big businesses of their own. **2021: Meesho removes commission.** Sellers paid nothing to list or sell. Meesho earned instead from shipping fees and ads, reached first-time buyers in small towns with ₹200 products, and by 2025–26 forced Flipkart and Amazon to drop commission on items under ₹1,000. **2024–26: ten-minute delivery.** Blinkit, Zepto, Instamart, Flipkart Minutes and Amazon Now began taking the everyday, frequent purchases (phone chargers, beauty, even kurtas on Myntra’s M-Now) that used to keep shoppers opening the big apps.



Figure 3.6 · Twenty years of Indian e-commerce. The three red points changed how everyone earns: rules that forced the marketplace model, a rival that made selling free, and a speed race that moved frequent purchases elsewhere. *Company announcements; DPIIT Press Notes 3 (2016) and 2 (2018).*

Five mechanics that aren’t obvious

- “Zero commission” is not free selling.** Meesho, and now Flipkart and Amazon under ₹1,000, charge sellers through shipping fees and ads instead. A seller who wants to be seen on page one pays for it.
- Ads are the profit engine.** An ad costs the platform almost nothing to show, so most of the ad money is profit. Myntra’s ₹915 Cr of ads and Amazon’s \$19.8 bn a quarter explain most of their margins. The risk: too many ads push relevant products down and conversion falls.
- Search ranking is the shelf.** Most buyers never scroll past the first screen of results. Deciding what shows in the top four slots is where the platform balances relevance, seller fairness and ad revenue.
- A sale week is a stress test.** Big sale events bring a large share of the year’s phone and appliance sales into a few days. The site must not fall over at midnight, bank offers must be ready, and sellers must have stock in the right

warehouses weeks in advance.

5. **Cash on delivery buys trust and sells risk.** First-time buyers in smaller towns trust COD, so it wins them. But a COD buyer can refuse at the door for free, which drives the RTO cost. Platforms nudge towards UPI with small discounts or a fee on COD.

First screen for "cotton kurta women"

1 • Sponsored

won an auction: bid × how well it sells

2 • Organic

earned: relevance and past buy-and-keep

3 • Organic

earned

4 • Sponsored

fixed ad slot in every screen

What earns an organic slot (typical signals)

- ① Matches the words: "cotton", "kurta", "women"
- ② Sells and is kept when shown: past conversion, return rate
- ③ Can arrive fast: stock in a nearby warehouse
- ④ Priced well against similar kurtas
- ⑤ Seller is reliable: ratings, cancellations

The PM's dial: every extra ad slot earns money now and costs conversion if relevance drops.

Page 5 exists, but few shoppers ever get there.

Figure 3.7 • Search ranking is the shelf. Sellers reach the first screen in two ways: pay for a sponsored slot, or earn an organic one by selling well and being kept. How many slots go to ads is one of the biggest product decisions on a shopping app. *Simplified; signals and slot layout vary by platform and change often.*

REAL-LIFE EXAMPLE • WHY MEESHO GREW WHILE CHARGING SELLERS NOTHING

A seller in Surat makes ₹250 sarees. On a platform charging 20% commission, ₹50 of each sale goes in fees, which kills the margin. On Meesho the seller pays no commission and only the shipping. So thousands of small sellers listed there, prices stayed low, and buyers in small towns found products at prices they had never seen online. Meesho earns from shipping margins and seller ads. By FY26 it had 264 million buyers and had cut its loss by two-thirds.

IN THE INTERVIEW • QUESTIONS THIS SECTOR LOVES

- "Return rates are rising. What would you do?" Split the rise into category mix and within-category rates before proposing anything.
- "How would you price Prime?" or "Should Amazon cut the Prime fee?" Cost floor, value ceiling, and how many new members a cut must bring in.
- "Design a feature using AI for a shopping app." First ask whether the problem needs AI at all.
- "How do you measure the health of a marketplace?" Talk about both sides and whether they meet.
- "Should Flipkart copy Meesho?" and "How does quick commerce threaten Amazon?" Who pays whom, and which purchases move.

Try it first. Think of the last three times you opened a fashion app without buying anything. What brought you there? Which of those reasons would a chatbot make better, and which would a simple rule fix?

THE QUESTION

You are a Product Manager at Myntra. The leadership believes that Artificial Intelligence (AI) is the key to the next phase of growth. How would you leverage AI to significantly increase customer engagement on the app?

A Interesting. Before I dive in, I'd like to clarify the scope. When we say "leverage AI," are we looking at Generative AI (like chatbots/image generation), Predictive AI (recommendations), or Computer Vision? Or am I free to explore the best fit?

Q You are free to explore the best fit, but the primary goal remains increasing engagement (daily active usage and session time), not just transaction volume

A Got it. Myntra is already strong in transactional commerce. To drive engagement using AI, we need to solve problems that keep users coming back even when they aren't ready to buy immediately.

I will follow this structure:

1. User Personas & AI-Specific Pain Points
2. AI-Driven Solutions
3. Prioritization (Impact vs. AI Feasibility)
4. Success Metrics

Q Sounds good. Let's start with the personas

A I'll focus on three personas where AI can bridge the gap between "browsing" and "experiencing":

The "Clueless" Dresser (Male/Female, 20-35): Has a wardrobe but struggles to match items. Often thinks, "I have this skirt, but what do I wear with it?" They need a Stylist.

The Trend Hunter (Gen Z, 18-25): Sees outfits on Instagram/Pinterest or in movies and wants that exact look immediately. They suffer from Discovery Friction.

The Fit-Anxious Shopper: Hesitates to browse because they can't visualize how clothes look on their specific body type. They need Validation.

I will prioritize The "Clueless" Dresser.

Reason: This user has high intent but low confidence. If we use AI to solve their "styling" problem using their existing purchases, we create a daily use case ("What do I wear today?") rather than just a shopping use case.

Q That's a strong choice. Solving the "post-purchase" or "pre-wear" anxiety is a great engagement hook. What AI solutions do you propose?

A I propose three AI-centric features:

"Myntra Mirror" (GenAI Virtual Studio): A Generative AI feature where users upload a selfie. The AI generates a hyper-realistic image of them wearing any product.

The AI Tech: Generative Adversarial Networks (GANs) for image synthesis.

Engagement Hook: Users spend hours "trying on" expensive brands or experimental styles just for fun/social sharing, increasing session time.

"StyleGPT" (Conversational AI Stylist): An in-app chatbot powered by LLMs (Large Language Models) trained on fashion trends.

Use Case: User asks, "I have a brunch date in Bangalore, it's raining. Suggest an outfit from my past purchases + one new item."

The AI Tech: LLMs + RAG (Retrieval-Augmented Generation) on Myntra's catalog.

Engagement Hook: High interactivity; users treat the app like a consulting friend.

"Snap & Shop" (Visual Search 2.0): Users upload a screenshot of a celebrity or an influencer from Instagram. The AI analyzes the fabric, cut, and pattern to find the exact or closest match in Myntra's catalog instantly.

The AI Tech: Computer Vision & Deep Learning.

Engagement Hook: Reduces the friction of "searching for a red floral dress" to just uploading a photo.

Q These are all viable. Which one would you build first and why?

A I would prioritize "StyleGPT" (The AI Stylist).

Feature	Impact (Engagement)	AI Complexity/Cost	Priority
Myntra Mirror (Try-On)	High (Viral potential)	High (High compute cost, latency issues)	P2
StyleGPT (Chatbot)	High (Daily utility, personalization)	Medium (APIs exist, fine-tuning needed)	P1
Snap & Shop (Visual)	Medium (Transactional utility)	Medium (Commoditized tech)	P3

A StyleGPT is my P1 because:

Retention: It utilizes the user's past order history, creating a lock-in effect (data moat). No other platform knows what you already own.

Feasibility: Text-based LLM integration is currently more stable and cost-effective than real-time video/image generation (Virtual Try-On).

Conversational Engagement: Chatting increases session duration significantly compared to just scrolling.

Q Fair point. Text-based AI is easier to ship quickly. What metrics would you track to ensure this actually drives value?

A North Star Metric: # of AI Interactions per DAU (How many users are actually talking to the stylist daily).

L1 Metrics:

Turn-to-Product Ratio: How many chat conversations lead to a product page view?

Session Duration: Pre-launch vs. Post-launch of StyleGPT.

Retention: D30 retention of users who engaged with StyleGPT vs. those who didn't.

Counter Metric: Hallucination Rate / Bad Recommendations: We need to measure "Thumbs Down" on AI suggestions to ensure we aren't recommending winter coats for summer, which would break trust.

Q Excellent. This aligns perfectly with our AI strategy. We can close the case.

WHAT WORKED · WHAT WAS MISSING

- ✓ **Asked what "AI" meant** and got a precise goal back: more days of use and longer sessions, not more orders.
- ✓ **Picked one persona with a reason:** a "what do I wear today?" need can bring people back daily, a shopping need can't.
- ✓ **Named the tech and the hook for each idea**, and ranked them on impact against cost.
- ✓ **Chose text first because it is cheaper and more reliable**, and added a counter metric for bad suggestions.
- ✗ **Accepted "AI is the answer" without testing it.** Fit anxiety, for one, is mostly solved by data and a rule ("62% of buyers chose one size up"), with no model at all.
- ✗ **Never sized the problem.** How many users open the app only when buying? How many have enough Myntra purchases to style?
- ✗ **No cost.** An LLM chat has a price per message; at millions of users a day that decides whether the feature survives.
- ✗ **The North Star rewards chattiness.** A stylist that confuses people gets more messages, not fewer.
- ✗ **Didn't ask what Myntra already has.** It launched MyFashionGPT (search by typing a need in plain words) in 2023 and has visual search.

Scoping kit check: Confirm the thing ✓ (**which kind of AI**) · Goal ✓ · Who ✓ · Today and the trigger: what Myntra already ships ✗ · Edges: budget, platform, timeline ✗. Skipping "today" is why the answer proposed a chatbot Myntra already has a version of; skipping edges is why cost never came up.

CHECK · "NO OTHER PLATFORM KNOWS WHAT YOU ALREADY OWN" **OVERSTATED**

Myntra knows only what you bought *on Myntra*. Most Indian wardrobes are bought in shops, as gifts and on other apps. A user with three Myntra orders has too few items for a daily outfit. The data moat is real only for heavy buyers, or if users add photos of the rest of their clothes.

CHECK · "CHATTING INCREASES SESSION DURATION SIGNIFICANTLY" **OVERSTATED**

Longer sessions can mean people can't find what they want. What matters is whether they come back on more days and act on the suggestion. Many apps have seen chat features get a burst of curiosity in week one and then fade.

STRONGER ANSWER



Figure 3.8 · An AI-feature answer in six moves. Move ③ is the one most candidates skip: check whether the problem needs a model at all before choosing one. *Author's summary.*

A ① **CLARIFY**

"Engagement" can mean two things: people open Myntra on more days, or they stay longer each time. They lead to different ideas, so I'd ask which one matters. I'd also ask which users we care about, and what AI Myntra already has, such as visual search and MyFashionGPT, so I don't propose something that exists. I'll assume a mobile launch within six months.

Q More days a week. Focus on 18–30-year-olds. Assume visual search exists.

Why this move: “days a week” rules out ideas that only make each visit longer, such as a chat that keeps you typing. It also tells us what to measure at the end.

A ② SIZE IT

Most people open a fashion app only when they need to buy something. Illustrative: of 100 monthly users, about 60 come only when buying, a few times a year. Short interviews would give three reasons they don’t come on other days: nothing new that suits me, I don’t know what goes with what, and I’m scared of ordering the wrong size. Only the second reason can create a habit on days you are *not* buying, the way people check the weather every morning. The other two matter when you are already shopping.

Why this move: fixing the reason that creates a daily need moves “days a week”; fixing the others mostly moves sales.

A ③ DOES IT NEED AI?

Take each problem on its own. **Fear of size** doesn’t need a model: our returns data can say “runs small, most people chose one size up”, and that is a simple rule. **Styling** does: picking which bottoms and shoes go with a mustard kurta is a judgement nobody can write down as rules, we have millions of saved and bought outfits to learn from, and a bad suggestion costs the user only a swipe (Figure 3.9).

Why this move: it puts AI only where rules fail and mistakes are cheap, and ships the cheap fix for size straight away.

A ④ PICK ONE, CHECK FOUR RISKS

The idea: a daily “outfit of the day” built from what you own. Its four risks are in Figure 3.10. The biggest is **data**: most users own too few Myntra items to style, because they bought most of their clothes elsewhere. So the first version asks users to add photos of three favourite pieces.

Cost, illustrative: 10 lakh users a day × 3 outfits each × ₹0.20 of model cost per outfit = ₹6 lakh a day. ₹6 lakh × 365 days ≈ ₹22 Cr a year. The feature must bring in more than that in extra purchases.

Why this move: if the riskiest part fails, nothing else matters, so it is tested first and cheaply.

A ⑤ MVP

A morning card, not a chatbot: “Today: your blue jeans + this kurta + white sneakers”, with three buttons: Wear it, Swap, and Shop the missing piece. Launch it to 5% of users who own five or more Myntra items. Keep a similar group without it (a holdout) so we can see what the card really changes.

Why this move: a card needs no typing and costs one model call a day; a chat invites long, costly conversations. The holdout separates the card’s effect from everything else happening that month.

A ⑥ MEASURE

Goal (North Star): weekly users who save, wear or buy from a suggested outfit. **Inputs:** share of users with enough wardrobe items; outfits accepted per 100 shown. **Safety checks (guardrails):** return rate on items bought from suggestions; thumbs-down rate; model cost per active user. **Trade-off:** the card is cheap but narrow; chat is flexible but costly. **Next time:** open chat only to people who already use the card every week.

Why this move: the goal counts people who act on an outfit, so a confusing stylist can’t score well just by generating more chat.

HOW THE ANSWER HANGS TOGETHER

The goal (people acting on outfits each week) comes straight from the clarified meaning of engagement (more days a week). The biggest risk (too little wardrobe data) shaped the MVP (add three photos). The cost maths (₹22 Cr a year) set the bar the feature must clear. And the guardrail (returns on suggested items) stops an AI that sells clothes people send back.

FRAMEWORK · DOES THIS NEED AI?

1 · The situation. Leadership says “use AI”. Some of the user problems here need a model; others need a rule and a good label. Building a model where a rule works costs months and money, and adds wrong answers.

2 · An everyday example. A chai stall owner wonders whether to buy software that predicts how much milk to order. If he sells about 200 cups every day, a line in his notebook (“order 20 litres”) is enough. Only if sales swing with rain, cricket matches and festivals, and he has months of past sales to learn from, would a prediction beat his notebook. Even then, a wrong guess costs him only some spilt milk.

Question	Chai stall milk order	Myntra: fit anxiety	Myntra: styling
Can a simple rule do it?	Yes, on steady days	Mostly: “runs small” from returns data	No: too many combinations
Is there data with right answers?	Months of sales	Returns tagged “too small”	Millions of saved and bought looks
Is a wrong answer cheap?	Some spilt milk	No: a return costs ~₹250	Yes: the user swipes
Verdict	Notebook	Rule + label first	AI, with an easy “no”

3 · The framework. Ask four questions in order (Figure 3.9). **Can a rule do it?** If a person can write the logic down, write it down. **Is there enough data with right answers?** (Called **labels**: past examples where we know the right answer, like “returned: too small”.) **How costly is a wrong answer?** Cheap mistakes suit AI; costly ones need a human check. **Does it beat the simple version by enough to pay for itself?**

Using it, step by step (on this case):

1. List the user problems separately: fear of the wrong size, not knowing what goes together, nothing new that suits me.

2. For each one, try to write the logic as a single rule. “Tell people when a kurta runs small” fits in one sentence, so ship the rule.
3. Where no rule fits (styling), check you have past examples with known right answers: outfits people saved, bought and kept.
4. Ask what a wrong answer costs the user. A bad outfit costs a swipe, so AI can decide alone; blocking a seller costs a livelihood, so a person checks.
5. Test the model against the simple version on a small group. Keep it only if it wins by more than its running cost.

COMMON SLIP · judging the whole idea (“an AI stylist”) instead of each problem inside it. Most ideas contain one part that needs AI and two that don’t.

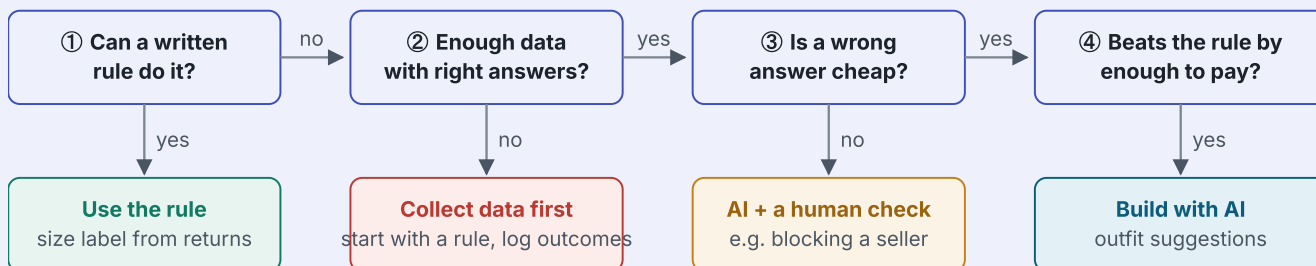


Figure 3.9 · Four questions before building with AI. Most ideas leave at the first box: if a person can write the logic down, a rule is cheaper, faster and easier to explain. AI earns its cost only at the far right. *Author’s summary, adapted from the “framework for applying AI/ML” in IIM A Product Mindset 2026–27, p. 89.*

4 · When it misleads. A rule that looks simple can quietly grow into hundreds of exceptions. “Brand X runs small” works until Brand X changes its factory mid-season and half its kurtas now run large; the rule keeps telling people to size up, and returns jump. Re-ask question ① whenever a rule needs a new exception every week.

5 · In the interview. “Before choosing a model, I’ll check whether a simple rule gets us most of the way. I’d use AI only where rules fail and a wrong answer is cheap.”

FRAMEWORK · THE FOUR RISKS OF A NEW PRODUCT (MARTY CAGAN)

1 · The situation. The candidate ranked ideas on “impact” and “AI complexity”. That misses the ways a product can fail even when it is easy to build: nobody wants it, people can’t figure it out, or it loses money.

2 · An everyday example. Your aunt wants to start a home tiffin service. Four things can sink it. Will office-goers nearby want home food at ₹120? Can they order and pay easily on WhatsApp? Can she cook 50 tiffins by noon in her kitchen? And does ₹120 cover vegetables, gas, packing and delivery, with the housing society’s permission?

Risk	Tiffin service	Myntra outfit of the day	Cheapest test
Value: will they want it?	Office-goers pay ₹120 for home food?	Will people open the app for an outfit on non-buying days?	Stylists pick outfits by hand for 200 users for two weeks
Usability: can they use it?	Ordering on WhatsApp	Do people understand Swap and Wear it without typing?	Five-user test of a clickable prototype
Feasibility: can we build it?	50 tiffins by noon	Enough wardrobe data; good outfits from few items	Run the model on 1,000 real wardrobes, rate the output
Viability: does it work for the business?	₹120 covers costs; society allows it	₹22 Cr a year of model cost pays back in purchases; no brand conflicts	Spreadsheet: cost per user vs extra purchases per user

3 · The framework. Marty Cagan (SVP, *Inspired*) says every idea carries four risks: **value** (will customers want it?), **usability** (can they work out how to use it?), **feasibility** (can we build it with our time, skills and data?) and **viability** (does it work for the business: cost, law, brand, partners?). The point is to test the riskiest one first and cheaply, before building (Figure 3.10).

Using it, step by step (on this case):

1. Write the idea in one line: a daily outfit card built from what the user owns.
2. For each of the four risks, write the one thing that would kill it. Value: nobody opens it. Usability: people don’t get the buttons. Feasibility: too few wardrobe items. Viability: model cost beats extra sales.
3. Mark the riskiest: the one you know least about *and* that would hurt most. Here, wardrobe data.
4. Design the cheapest test for that one: run the model on 1,000 real wardrobes and have stylists rate the outfits.
5. Only when that passes, move to the next risk, and only then design the screens.

COMMON SLIP · testing usability first because designing screens is the fun part, while the real unknown is whether the data or the demand exists.

Value: will they want it?

Do people open Myntra on non-buying days for an outfit? Test: human stylists, 200 users.

risk: medium

Usability: can they use it?

A card with three taps, no typing.
Test: five users, one clickable prototype.

risk: low

Feasibility: can we build it?

Most users own too few Myntra items to style.
Test: run it on 1,000 real wardrobes.

risk: HIGH, test first

Viability: does it pay?

~₹22 Cr a year of model cost (illustrative)
must come back as extra purchases.

risk: medium

Figure 3.10 • The four risks, applied to an outfit-of-the-day card. The red box is the one to test first, because if it fails nothing else matters.

Framework: Marty Cagan, *Inspired* (2nd ed., 2017); ratings and costs illustrative.

4 • When it misleads. The four risks pull against each other. Cutting model cost (viability) by switching to a smaller model can make outfits worse (value). Google Glass passed feasibility in 2013 and failed value and social acceptance: people didn't want a camera on a stranger's face. Ticking each box separately misses the trade-off between them.

5 • In the interview. "This idea is easy to use but hard to feed with data, so I'd test the data risk first, cheaply, before designing screens."

CASE 3.2 • MEASURING FACEBOOK MARKETPLACE

Metrics

Medium IIM A Product Mindset 2026–27, p. 290

Try it first. You list an old study table on Marketplace for ₹2,000. What would make you say "this app works"? Now answer as the person who wants to buy a table. Is it the same thing?

THE QUESTION

You are the PM for Facebook Marketplace. What metrics would you track to measure the health and success of the feature?

A Before jumping into metrics, I'd like to confirm my understanding. Facebook Marketplace lets buyers and sellers discover, buy, and sell products within their local communities. Sellers can list both new and second-hand products, while buyers discover listings through Marketplace or Facebook groups. I'll assume the primary business goal is to increase user engagement on the platform.

Q You are right. Please go ahead.

A Before defining metrics, I'd like to map the journeys of both stakeholders so the metrics align with the user actions.

Seller Journey

- Visit Marketplace
- Create a listing
- Add product details & price
- Receive buyer interest
- Accept/reject an offer
- Receive payment
- Handover the product

The last two steps happen outside Facebook's ecosystem.

Buyer Journey

- Visit Marketplace
- Search/discover a product
- Contact the seller
- Make payment
- Receive the product

There is also another entry point where buyers discover listings through Facebook Groups before reaching Marketplace.

Q Your understanding of the journey of the stakeholders is fair. Let's move ahead to the metrics

A I'd group the metrics into four buckets: Awareness, Activation, Conversion and Retention.

1. Awareness: Measures whether users are discovering Marketplace.

- Users visiting Marketplace
- Marketplace tab clicks/day

2. Activation: Measures active participation on the platform.

- Products listed/week
- Bounce rate

- Product listing clicks/week
3. Conversion: Measures successful buyer-seller interactions.
- Unique interactions/week
 - Purchases/week
4. Retention: Measures repeat usage by buyers and sellers.
- Repeat purchase rate
 - Repeat listing rate

Q Alright, out of all these metrics mentioned, which one should be prioritized by us as the top metric?

A I'd segment Activation and Conversion metrics by product category to uncover category-specific trends. Since the objective is increasing engagement, my North Star Metric would be:

Unique Interactions per Week This captures meaningful buyer-seller activity on the platform.

As a counter metric, I'd track:

Purchases per Week This ensures we're not optimizing engagement at the cost of successful transactions or marketplace quality.

WHAT WORKED · WHAT WAS MISSING

- ✓ **Restated the product** in one line and stated the goal as an assumption, which the interviewer accepted.
- ✓ **Mapped both journeys**, buyer and seller, before any metric, and spotted that payment and handover happen off Facebook.
- ✓ **Grouped metrics into stages**, then chose one North Star and one counter metric when pushed.
- ✓ **Segmented by category**: furniture and phones behave differently.
- ✗ **Never tested whether buyers and sellers actually meet.** A marketplace is healthy when listings sell and searches find something. That idea, liquidity, is the heart of the case and never appears.
- ✗ **The counter metric can't be measured.** The candidate said payment happens outside Facebook, then chose purchases per week to guard the North Star.
- ✗ **No trust metric.** Scams and fake listings are Marketplace's best-known problem; "interactions" rise when scammers message people.
- ✗ **No local view.** A table in Pune only matters to buyers in Pune. City-level health is the real unit.

Scoping kit check: Confirm the thing ✓ · Goal ✓ (assumed, then confirmed) · Who ✓ both sides · Today and the trigger: mature product, about a billion monthly users ✗ · Edges: which markets ✗. Metrics add-on "what decision will the metric drive?" ✗: without it, the list grew to eleven metrics and no priority.

CHECK · "COUNTER METRIC: PURCHASES PER WEEK" DOESN'T HOLD UP

Most Marketplace deals are paid in cash or by UPI at the door, so Facebook can't see them. The observable proxies are a seller tapping "Mark as sold", a buyer rating the seller, and a conversation that ends with both sides agreeing a time. Pick metrics you can actually count.

CHECK · "UNIQUE INTERACTIONS PER WEEK" AS THE NORTH STAR OVERSTATED

Messages go up when buyers haggle endlessly, when a listing is misleading ("is this still available?" asked 40 times) and when scammers write to sellers. More interactions can mean a worse marketplace. The North Star should count *completed* matches.

STRONGER ANSWER



Figure 3.11 · A marketplace-metrics answer in six moves. Move ③ is what separates a marketplace answer from a generic app answer: check that the two sides meet. *Author's summary.*

A ① CLARIFY

Marketplace is mature, with about a billion monthly users, so this is about health, not launch. Two questions change the answer. What decision will these metrics drive: where to invest, or spotting problems early? And is the goal more time on Facebook, or Marketplace working as a place to buy and sell?

Q A monthly health review for the Marketplace team. Buying and selling that works; engagement follows.

Why this move: a monthly review needs a handful of numbers that point to action, not a list of eleven. And "buying and selling that works" tells us to measure deals, not clicks.

A ② MAP

There are two sides, and each has one moment that decides whether they come back. For the seller: did my table sell, quickly, at a fair price? For the buyer: did I find a table near me, from someone I could trust? Every other metric is a cause of one of these two moments.

Why this move: naming the moment of truth for each side keeps the metrics tied to what users actually care about.

A ③ LIQUIDITY

So the core question is whether listings find buyers. Illustrative, for 100 furniture listings in Pune in a week: 62 get a message within a day, 41 lead to an agreed meeting, and 30 are marked sold within 7 days (Figure 3.12). That 30% is the **sell-through rate**. I'd track it for each city and category, because a buyer in Pune can't use a table in Patna.

| Why this move: a marketplace is only as good as its chance of a match. Visits and messages are causes or side effects of that.

A ④ GOAL

Goal (North Star): weekly successful local matches, meaning listings marked sold after a Marketplace conversation, or confirmed by the buyer's rating. Payment happens off Facebook, so I pick signals we can actually see.

| Why this move: one number that both sides' success feeds into, and that Facebook can count.

A ⑤ INPUTS

I'd split the levers by side so each team knows what to move. **Sellers:** new listings a week from active sellers; share of listings with a clear photo and a price close to similar items. **Buyers:** share of searches that show 10 or more nearby listings; share of searches that lead to a message. **Between them:** median time to first reply; share of first messages answered within an hour.

| Why this move: when the goal drops, these tell you which side, and which step, to fix.

A ⑥ GUARDRAILS

Safety checks (guardrails): scam reports per 1,000 conversations; share of listings removed for breaking rules; how often sellers block buyers. **Trade-off:** stricter checks on new sellers cut scams but also cut listings, which lowers liquidity. **Next time:** show "thin" cities separately from "thick" ones in every chart.

| Why this move: scams and spam can inflate activity numbers, so the guardrails make sure growth is healthy growth.

HOW THE ANSWER HANGS TOGETHER

The decision (a monthly health review) told us to keep the list short. Mapping both sides gave two moments of truth, and liquidity joined them into one number: completed local matches. The inputs are split by side, so a drop points to a team. The guardrails stop scams from making the numbers look better than the marketplace really is.

FRAMEWORK · MARKETPLACE LIQUIDITY

1 · The situation. The candidate counted activity on each side, but nothing measured whether the sides met. A marketplace with many listings and many visitors can still fail, if the listings are in the wrong city or at the wrong price.

2 · An everyday example. A Sunday vegetable market (a *haat*). A vendor comes back next week only if most of her tomatoes sold before closing. A shopper comes back only if she found good tomatoes without walking the whole market. Count the vendors and shoppers and you learn little; count the share of tomatoes sold and the time shoppers took to find them, and you know if the market will survive.

Idea	Sunday vegetable market	Facebook Marketplace
Seller-side liquidity	Share of tomatoes sold before closing	Share of listings sold within 7 days (sell-through)
Buyer-side liquidity	Found good tomatoes in 10 minutes	Searches that end in a message to a seller
Speed	How long a vendor waits for the first customer	Time to first message; time to sale
Local	This market, this Sunday	This city, this category
Unhealthy but busy	Crowds that haggle and don't buy	Many messages, few sales

3 · The framework. **Liquidity** is the chance that a participant on one side finds a match on the other side within a reasonable time. Measure it from both ends: sell-through for sellers, search success for buyers, and the time each takes. Liquidity is local: it must be read city by city and category by category, because a buyer in Pune can't use a table in Patna. The side that is harder to attract (usually supply, the listings) gets the first investment.

Using it, step by step (on this case):

1. Name each side and what success means for it: a seller's table sells; a buyer finds a table nearby.
2. Pick one event you can see that proves a match happened: a listing marked sold after a Marketplace chat.
3. Build the funnel from listing to match (100 → 62 → 41 → 30) and find the biggest leak.
4. Cut the same funnel by city and category to find "thin" markets where few matches happen.
5. Invest in the harder side (usually listings) in the thinnest markets first.

COMMON SLIP · counting activity (visits, messages) as if it were matches. A busy market where nothing sells is a failing market.

100 listings

furniture, Pune, one week (illustrative)

62 get a

message within a day: buyers can find them

41

agree a time to meet: price and trust are OK

30

marked sold within 7 days: sell-through 30%

Where it leaks, and the fix

38 never get a message: bad photo, price far above similar tables

21 talk but don't meet: lowballers, no-shows, fear of scams

Figure 3.12 · Liquidity, read as a funnel from the seller's side. Only 30 of 100 tables sell in a week. The two leaks point to two different fixes: price and photo guidance for the first, trust signals and reminders for the second. *Illustrative numbers.*

4 · When it misleads. High sell-through can hide a price problem. If sellers price far below value (a new phone for half price), everything sells fast and sellers feel cheated later, or it is stolen goods. And a national average hides dead cities: 45% sell-through in Mumbai and 8% in small towns average to a healthy-looking 30%.

5 · In the interview. “A marketplace is healthy when the two sides find each other, so my North Star counts completed matches, and I read it city by city.”

CASE 3.3 · AMAZON'S RETURN RATE IS RISING

Root cause analysis

Medium IIM B *Sigma PM Casebook 2026–27*, pp. 70–71

Try it first. Your family's electricity bill in May is 40% higher than in February, yet no appliance has become less efficient. Explain how. Then use the same idea on a rising return rate.

THE QUESTION

There has been increase in the return-rate for Amazon products. What can be done?

A Sure. I have some initial clarifying questions before proceeding. Is the decline sudden or gradual? Does the decline occur during any particular season?

Q The increase has been gradual. But, the return-rate peaks during the wedding/festive season.

A Is there any specific product category for which the return-rate is increasing much?

Q Return rates of Clothes and Electronics have increased the most. You can focus on clothes for now.

A Okay. Is this problem prevalent across India or is it specific to some particular geographic locations?

Q We are facing this issue across India.

A Okay. I can think of 5 potential reasons for returning online ordered clothes–

1. **Wrong/defective products:** Sellers sometimes send defective or wrong item other than what the customer wants
2. **Wardrobing:** Sometimes customers buy clothes/footwear particularly for any occasion and then return the product after using it on that occasion.
3. **Size/fit issue:** It happens many times that the size of the product misfits the customer, so he/she has to return it.
4. **Change in mind:** Sometimes while searching for products, customers impulsively order items that they do not require. And after delivery, they realize that the product is not as appealing as they were anticipating and they have made an unnecessary purchase.
5. **Delay in delivery:** It happens sometimes that a customer requires something for a particular occasion, but the delivery gets delayed due to some reason and the customer no longer requires that product.

Is there anything else I need to add as a reason?

Q No, you have come up with good reasons. You can proceed.

A Should I focus on any particular reason?

Q You can focus on the first three issues. Suggest ways by which we can address these issues and minimize the return-rate.

A Sure. Those 3 reasons of return can be tackled by–

1. Wrong/defective products:
 1. Barcode authentication: Put a barcode not just on the shipping box, but on the product itself. And, scan the barcode at each stop during the transit and share the product details to the customer. This will alert the customer as soon as there is any change/defect in the product. It'll also help in tracking the package and minimize the chances of it getting lost/stolen or exchanged during transit.
2. Wardrobing:
 1. Reduce the return window for clothes and perform stricter checking (tamper-evident tags) during pickup of returned products. We can implement 15-minute return policy as a pilot.
 2. Flag customers with more than average return rate. Penalize the customer after a certain number of returns.
3. Size/fit issue:

1. Provide realistic product pictures, size-chart and insights from past purchasers. Incentivize reviews which help the customers.
2. 360 degree view and virtual try-ons.

Q How will you prioritize among these?

A For prioritizing features, I will look into the value added by the feature and the complexity of implementation for the company.

Feature	Value	Complexity
Barcode authentication	High	High
Reducing return window size	High	Medium - High
Flagging customers	High	Low
Realistic product pics and reviews	Medium	Medium
360 deg view and virtual try-on	Medium	Medium

Q These recommendations sounds reasonable. Pick your top 3 solutions.

A Sure. My top 3 recommendations would be:

1. Barcode authentication
2. Reducing return window size
3. Flagging customers

However, there can be dissatisfaction among customers due to reduction in return window size. We should implement it as a pilot for a few products and we have to be very vigilant about customers' reactions towards it.

Q Seems practical. That would be it. Thanks!

WHAT WORKED · WHAT WAS MISSING

- ✓ **Asked sudden or gradual, and which season.** "Gradual, with festive peaks" is the most useful fact in the case.
- ✓ **Narrowed by category and by region** before brainstorming.
- ✓ **A clean list of five reasons**, then fixes for each of the three chosen.
- ✓ **Saw the cost of a shorter return window** and proposed a pilot.
- ✗ **Never asked what the rate is, or by how much it rose.** From 18% to 20% is a different problem from 18% to 30%.
- ✗ **Missed the mix effect.** In the festive season far more clothes are sold, and clothes are always returned more. The total rate rises even if no category gets worse.
- ✗ **Guessed the reasons instead of reading them.** Every return has a reason code picked by the customer; the data would rank the five reasons in a minute.
- ✗ **The top pick is the costliest.** Barcodes on every product in transit fix "wrong or defective items", usually a small share of fashion returns, while size and fit is usually the largest.
- ✗ **Penalising customers hits good ones too.** A customer who orders two sizes and returns one is often a loyal, high-spending buyer.

Scoping kit check: Confirm the thing: which rate (orders or value, returns ÷ delivered)? ✗ · Goal ✗ · Who ✓ (**category**) · Today and the trigger ✓ (**gradual, festive peaks**) · Edges ✓ (**all India**). RCA add-ons: size vs baseline ✗ · where it shows ✓ · recent changes ✗ · is the data trustworthy ✗. The two skipped add-ons would have exposed the mix effect straight away.

CHECK · "WARDROBING" AS A TOP-THREE CAUSE **DIRECTIONALLY RIGHT**

Wearing an outfit once and returning it does happen, especially with party and wedding wear in the festive season. But industry surveys consistently put size and fit first by a wide margin. Wardrobing deserves a targeted fix (tamper tags on high-value ethnic wear), not a return-window cut for all clothes.

CHECK · "REDUCE THE RETURN WINDOW... A 15-MINUTE RETURN POLICY AS A PILOT" **OVERSTATED**

Easy returns are a reason people buy clothes online at all. Shrinking the window cuts returns partly by cutting sales. A gentler tool is try-and-buy at the door, which Myntra launched in 2017 as a ₹99 add-on: the customer tries the item while the rider waits, so a misfit costs one trip, not two.

STRONGER ANSWER



Figure 3.13 · A return-rate answer in six moves. Move ② decides whether there is a problem to fix at all. *Author's summary.*

A ① **CLARIFY**

I'd pin down three things. What exactly is the rate: returned units ÷ delivered units? From what level to what, over how long? And did anything change on our side: return rules, a new seller programme, a new delivery partner?

Q Units. Overall it went from 11% to about 15% over the festive quarter. No policy changes.

Why this move: “11% to 15% in a festive quarter, with no rule changes” already hints that the season, not a broken process, may be behind much of it.

A ② **MIX OR RATE?**

First I’d split the rise into two parts. Clothes are always returned more than other things, and in the wedding season we sell far more clothes. Illustrative: clothes went from 30% to 40% of orders. If every category kept its old return rate, that shift alone would add 2.1 of the 3.7 points. The rest comes from clothes’ own rate rising from 25% to 28% and electronics from 8% to 10% (Figure 3.14).

Why this move: more than half the “problem” is simply selling more clothes, which is good news. Only the smaller part is a real deterioration.

A ③ **REASON CODES**

For the part that is a real rise inside clothes, I’d read the reason each customer chose when returning, and cut it by brand, seller, size, payment type, and new versus repeat buyers. I’d also count **bracketing**: one order with the same item in two sizes, like Pooja’s kurta. Illustrative split of clothing returns: size and fit 55%, “didn’t like it” 20%, damaged or wrong item 10%.

Why this move: the data ranks the causes in minutes; guessing them risks spending effort on a small one.

A ④ **SIZE IT**

Each return loses about ₹250 (two trips, a check, a discounted resale). 3 extra points on, say, 1 crore clothing units a month = 3 lakh extra returns. $3 \text{ lakh} \times ₹250 = ₹7.5 \text{ Cr}$ a month. Size and fit is 55% of that, about ₹4 Cr, so it gets most of the effort.

Why this move: rupees turn a list of causes into a priority order anyone in the room can agree on.

A ⑤ **FIX BY ROOT CAUSE**

Size and fit: a fit hint on each product, built from its returns (“runs small: 60% chose one size up”), plus size advice based on what fitted the customer before. **Bracketing:** show that hint at the moment someone adds a second size.

Bad sellers: sellers whose return rate is twice their category’s get flagged and drop in search results. **Wardrobing:** tamper tags only on high-value party and ethnic wear.

Why this move: each fix targets one measured cause, so we can tell later which one worked.

A ⑥ **MEASURE**

Goal (North Star): kept sales per visitor (NMV per visit), not the return rate on its own. **Safety checks (guardrails):** conversion rate; complaints about refused returns. **Trade-off:** anything that makes returns harder lowers the return rate but also lowers sales. **Next time:** report festive-season return rates with the category mix held fixed, so the season stops looking like a crisis.

Why this move: a return rate can be “improved” by scaring buyers away. Kept sales per visitor can only rise if customers buy and keep more.

HOW THE ANSWER HANGS TOGETHER

Clarifying the size of the rise (11% to 15%) made it worth splitting. The mix-or-rate split showed that only about a third is a real deterioration. Reason codes and rupees then ranked what is left, and the fixes follow those causes. The goal, kept sales per visitor, makes sure we cut returns without cutting sales.

FRAMEWORK · MIX OR RATE? SPLITTING A CHANGE IN AN AVERAGE

1 · The situation. An overall rate is an average across groups: here, categories. It can change because the groups changed, or because the mix of groups changed. The candidate went straight to the groups.

2 · An everyday example. Your electricity bill is 40% higher in May than in February. The fridge, fans and TV use the same power per hour as before. What changed is the *mix of hours*: the air conditioner, which uses the most power, now runs six hours a night. No appliance got worse; the bill still went up.

Idea	Electricity bill in May	Amazon’s return rate in the festive season
Groups	Appliances	Categories: clothes, electronics, the rest
Mix	Share of hours each appliance runs	Share of orders in each category
Rate	Power per hour of each appliance	Return rate of each category
Mix effect	AC runs more hours	More clothes sold in the wedding season
Rate effect	An old fridge starts drawing more	Clothes returned more often than before

3 · The framework. Change in the overall rate = **mix effect** (how much the shares moved, at the old rates) + **rate effect** (how much each group’s rate moved, at the old shares) + a small bit from both moving together. Worked numbers in Figure 3.14: clothes rise from 30% to 40% of orders at a 25% return rate, so the mix effect for clothes is $10 \text{ points} \times 25\% = 2.5$ points; the rest of the catalogue shrinks by 10 points at 4%, giving back 0.4. Mix: +2.1. Rate: clothes $30\% \times 3 \text{ points} = 0.9$, electronics $20\% \times 2 \text{ points} = 0.4$. Rate: +1.3. Both moving together: $10\% \times 3 \text{ points} = 0.3$. Total: 3.7 points.

Using it, step by step (on this case):

1. Pick the groups (categories) and write down each group’s share of orders and return rate, before and after.
2. Mix effect: for each group, change in share $\times$ old rate. Add them up ($+2.5 - 0.4 = +2.1$).
3. Rate effect: for each group, old share $\times$ change in rate. Add them up ($+0.9 + 0.4 = +1.3$).

- The small remainder is both moving at once (+0.3). Check the three parts add to the total change (3.7).
- Explain the mix part as a business change (more clothes sold); investigate only the rate part.

COMMON SLIP · mixing old and new numbers inconsistently, which double counts. Always use old rates for the mix effect and old shares for the rate effect.

Normal quarter: overall 11.1%

Clothes 30% returned 25%	Elec. 20% 8%	Everything else 50% returned 4%
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Festive quarter: overall 14.8%

Clothes 40% returned 28%	Elec. 20% 10%	Everything else 40% returned 4%
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bar width = share of orders; every clothing and electronics rate is higher, but the width of the orange bar matters more

Where the 3.7 extra points come from

Mix: +2.1 points	more clothes sold
Rate: +1.3	clothes, electronics returned more often
	both together: +0.3

Only the red bar (about a third) is a real deterioration to investigate.

Figure 3.14 · Splitting a rising return rate into mix and rate. Selling more clothes in the wedding season explains more than half the rise, even before any category gets worse. *Illustrative numbers built to match the case: overall 11% to about 15%.*

4 · When it misleads. The answer depends on how you group. “Clothes” may look steady while kurtas soar and T-shirts fall, so split one level deeper where the money is. And mix can reverse a conclusion entirely (called Simpson’s paradox): a seller can have a lower return rate than a rival in every single category and still a higher rate overall, just because it sells more clothes.

5 · In the interview. “Before looking for what got worse, I’ll check how much of the rise is just the mix shifting towards clothes. Only the part left over needs a fix.”

CASE 3.4 · PRICING AMAZON PRIME, TWO TAKES

Pricing Medium IIM A p. 297 · IIM B p. 94

Try it first. Prime in India costs ₹1,499 a year. Your boss wants to cut it to ₹1,299. How many new members would the cut need to bring in just to break even? Write the one line of maths before reading on.

Two casebooks ask about the same product from opposite ends. **Take A** (IIM A *Product Mindset 2026–27*, p. 297) tests a proposed price cut in the US. **Take B** (IIM B *Sigma PM Casebook 2026–27*, p. 94) sets a price from scratch for India. Read them together: one is all maths and no method, the other all method and little maths.

TAKE A · SHOULD PRIME DROP FROM \$99.99 TO \$89.99?

THE QUESTION

Assume you are the new product manager in our Amazon Prime business and are deciding pricing. The vice president would like to lower the price from \$99.99 per year to \$89.99 per year. Making your own assumptions, develop the financial projections for this decision.

A A lower price should increase Prime adoption. At the same time, it will reduce subscription revenue from existing members and increase shipping costs. I’ll estimate the impact separately for existing and new Prime customers. Let’s assume there are 200 million Prime members today.

1. Existing Prime Customers

Reducing the membership fee by \$10 leads to a direct revenue loss of: $\$10 \times 200\text{M} = \2 billion

Net impact: -\$2B

2. New Prime Customers

Next, I’ll estimate the impact from new customers acquired due to the price reduction. Let’s assume Prime membership grows by 20%, adding 40 million new members.

Membership Revenue: Incremental membership revenue: $40\text{M} \times \$90 = \3.6B

Incremental Shipping Cost: Assume each new Prime customer increases shipping costs by \$20/year. $40\text{M} \times \$20 = \0.8B

Incremental Commerce Revenue: Prime members tend to purchase more because of free 2-day shipping. Assume each new Prime member spends an additional \$40/year. At a 25% gross margin, that’s \$10 profit/customer. $40\text{M} \times \$10 = \0.4B

To recap the annual impact:

Existing Prime customers: -\$2.0B

New Prime customers:

- +\$3.6B Membership revenue
- -\$0.8B Incremental shipping cost

- +\$0.4B Incremental gross margin

Net impact: +\$3.2B

Q So, what's the overall change in revenue, and what's your recommendation?

A Overall annual impact:

-\$2.0B + \$3.2B = +\$1.2B

Based on these assumptions, reducing the annual membership fee by \$10 results in an estimated \$1.2B net annual upside.

Recommendation: Proceed with the price reduction.

TAKE B · HOW WOULD YOU PRICE PRIME IN INDIA?

THE QUESTION

How would you price Amazon Prime?

A Just to be clear, Amazon prime is a bundled service that offers multiple benefits like Amazon Prime Shipping Benefits, Prime Deals and offers, Prime Video, Prime Music, Credit card rewards and Prime Reading. Am I correct with this?

Q Yes, you are correct.

A Are we focusing on any specific geography for this decision?

Q We are focusing on India.

A Got it! Are we aiming to maximize our market share, or are we focusing on maximizing our profits with this pricing strategy?

Q Our goal is to attract more customers and boost our market share.

A Got it! Then we should offer our service at a competitive price in order to penetrate the market and acquire customers. Can I go ahead with this assumption?

Q Yes, you can proceed.

A There are primarily three methods to come up with a price point.

- First is the Cost-based pricing model. This is based on the costs incurred by Amazon to provide the services.
- Second is the Value-based pricing model. This is based on the customer's perceived value of the services provided.
- Third is the competitor-based pricing model. In this model, the price charged by the competitors for similar service is used to arrive at the price. Since, Amazon Prime offers a bundle of services it is harder to find the cost incurred. So, I would like to use a combination of the Value-based and the Competitor-based models to arrive at the price. Can I go ahead with this method?

Q Sure, you can continue.

A With the help of the models, we see that:

- Prime users get free delivery on all their orders. A typical Prime user orders at least once a month, saving ₹50 per delivery, totaling Rs.600 per year.
- The direct competitors for our OTT service are Netflix and Disney+ Hotstar. They charge around ₹900 per year, giving us the same value.
- The direct competitor to Prime Music is Spotify, which costs around ₹1100 per year, giving us the same value.
- Prime reading is similar to Amazon Kindle, which costs around ₹1800 per year, giving us the same value.
- In addition to these Prime members also get exclusive deals, discounts and offers on credit cards, providing additional value to our customers. Thus, we arrive at a value of ₹4400. The industry standard is that the companies try to capture around 1/3rd of the customer's total perceived value. So, we should price our offering at around ₹1500, this should allow us to improve penetration and acquire market share.

Q That was a thorough analysis. I love the fact that you brought in so many numbers into the analysis. Thank you!

WHAT WORKED · WHAT WAS MISSING

- ✓ **Take A split existing and new members**, the right first cut for any price change, and counted shipping cost and extra shopping, not just fees.
- ✓ **Take A gave a number and a decision.**
- ✓ **Take B confirmed the bundle, the market and the goal** (market share) before pricing.
- ✓ **Take B named cost, value and competitor pricing**, explained why cost is hard for a bundle, and valued each benefit.
- ✓ **Take B's answer, about ₹1,500, is almost exactly the real price** (₹1,499 in 2026).
- ✗ **Take A's whole answer rests on one guess:** "membership grows 20%". It never asked how many new members the cut needs (12.5%, see below) or whether that is realistic.
- ✗ **Take A gives the cut to 200 million people who already pay.** That \$2 bn is the cost of the idea; a new-member offer avoids most of it.
- ✗ **Take B adds up every benefit as if each member uses all of them.** Most use one or two, so the average value is far below ₹4,400.

✗ **Neither tests before deciding.** Amazon can try a price on one group of new visitors for a month.

Scoping kit check (Take A): Confirm the thing ✗ · Goal: why does the VP want a cut? ✗ · Who: existing vs new ✓ · Today: current members, churn ✓ (**assumed 200 mn**) · Edges ✗. Pricing add-ons: share or profit ✗ · cost floor ✓ (**shipping**). **(Take B):** Confirm ✓ · Goal ✓ · Market ✓ · competitor prices ✓ · plan structure (Lite, Shopping Edition) ✗.

CHECK · “LET’S ASSUME PRIME MEMBERSHIP GROWS BY 20%” DOESN’T HOLD UP

A 10% price cut bringing 20% more members means demand is very sensitive to price. Prime’s history points the other way: Amazon raised the US price from \$99 to \$119 (2018) and \$139 (2022), and membership kept growing. People who haven’t joined at \$99 mostly don’t shop enough on Amazon, and \$10 won’t change that. The recommendation flips if growth is below 12.5%.

CHECK · “COMPANIES TRY TO CAPTURE AROUND 1/3 OF THE CUSTOMER’S PERCEIVED VALUE” OVERSTATED

There is no industry standard. The share of value a firm can charge depends on competition and on how sure customers are of the value. A rule of thumb is fine if you call it one; the stronger move is to check it against what similar bundles charge.

STRONGER ANSWER (TAKE A)



Figure 3.15 · A price-change answer in six moves. Move ④ turns a guess (“members grow 20%”) into a question the business can test (“will they grow 12.5%?”). Author’s summary.

A ① CLARIFY

Why does the VP want a cut: are sign-ups slowing, is a competitor cheaper, or do we want members to shop more? Is the goal more members or more profit? And would the new price apply to everyone, or only to people who join from now on?

Q Sign-ups have slowed. The VP wants growth, but not at a loss. Assume the cut applies to everyone.

Why this move: “growth, but not at a loss” turns the question into a sum: does the cut bring in enough new money to cover what it gives away?

A ② FLOOR

Start with the least we could charge. Each new member costs about \$20 a year in extra free shipping, plus a share of the video catalogue. \$89.99 is far above that, so we won’t lose money on any single new member. Cost isn’t what decides this.

Why this move: checking the floor first rules out one worry quickly, so the rest of the answer can focus on the real risk.

A ③ CEILING

Now the most members would pay. Existing members already pay \$99.99 and stayed through two earlier price rises, so Prime is worth more than that to them. They don’t need a cut to stay. For them, the cut is simply money given away: $\$10 \times 200 \text{ million members} = \2 bn a year .

Why this move: it finds where the cost of the idea really sits: in the discount handed to people who would have paid anyway.

A ④ BREAK-EVEN

How many new members pay back that \$2 bn? Each new member brings the \$90 fee, minus \$20 of shipping, plus about \$10 of margin on the extra shopping members do: $\$90 - \$20 + \$10 = \80 a year. $\$2 \text{ bn} \div \$80 = 25 \text{ million new members}$. On a base of 200 million, that is 12.5% growth (Figure 3.16). So a 10% price cut must bring 12.5% more members: every 1% off the price must add 1.25% more members.

Why this move: it replaces a guess (“members grow 20%”) with a threshold (“we need 12.5%”) that a test can confirm or reject.

A ⑤ TEST

Before any national change, show \$89.99 to a random 5% of people who aren’t members, for six weeks, and compare sign-ups with everyone else. If sign-ups rise less than 12.5%, drop the idea. A cheaper test: a first-year price for new members only. It keeps most of the \$2 bn, because existing members keep paying \$99.99.

Why this move: six weeks of real behaviour is worth more than any assumption, and the new-member version limits the downside.

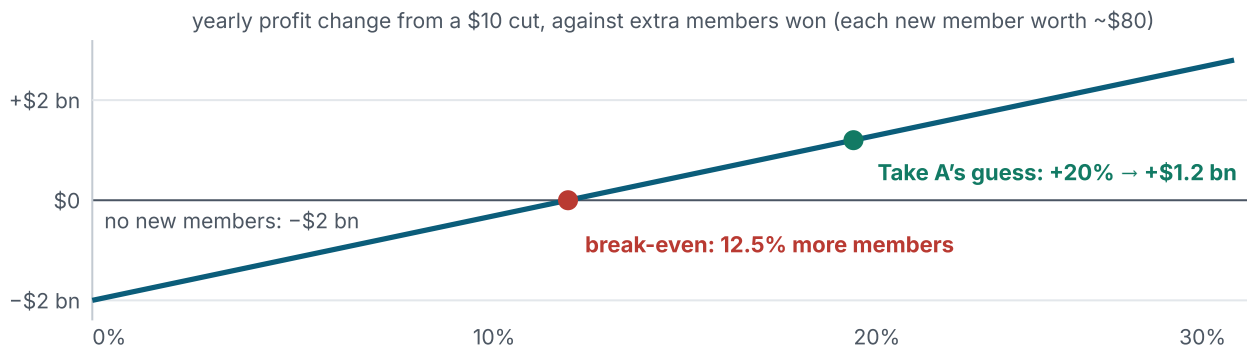


Figure 3.16 · How many new members does a \$10 cut need? Below 12.5% growth the cut loses money; Take A's 20% sits comfortably above the line, which is why the whole recommendation rests on that one assumption. *Take A's own numbers: 200 mn members, \$20 shipping and \$10 margin per new member.*

A ⑥ DECIDE

My recommendation: don't cut the price for everyone; test a new-member price. **Goal (North Star):** Prime profit, from fees plus the extra shopping members do. **Safety checks (guardrails):** how many existing members cancel; how much new members shop (bargain joiners who never shop are a loss). **Trade-off:** a new-member deal can annoy loyal members who notice it.

Why this move: the decision follows the numbers: the only part of the idea that can pay is the new-member part, so that is the part to test.

HOW THE ANSWER HANGS TOGETHER

The floor showed each member is profitable at either price. The ceiling showed existing members don't need the cut, so it costs \$2 bn. Break-even turned that into a target (25 million new members, 12.5%), and the test checks the target cheaply. The recommendation keeps the upside (new members) and drops the costly part (discounting loyal ones).

FRAMEWORK · THREE WAYS TO SET A PRICE, AND THE BREAK-EVEN TEST FOR A CUT

1 · The situation. Take B named three pricing methods; Take A needed a fourth tool, a break-even test, and didn't use it. Together they cover the two pricing questions interviewers ask: "what should it cost?" and "should we change the price?"

2 · An everyday example. Your friend bakes chocolate cakes at home. Ingredients and gas cost ₹300 a cake: below that she loses money. The bakery down the road sells a similar cake for ₹600. A family ordering for a birthday would happily pay ₹900 for hers, because it's better and delivered warm. So her price sits between ₹600 and ₹900. If she thinks of cutting from ₹700 to ₹600, she should first ask how many extra cakes that must sell to make up the ₹100 lost on every cake she already sells.

Idea	Home-baked cake	Amazon Prime India
Cost floor	₹300 of ingredients and gas	Cost of free deliveries and the video catalogue per member
Competitor anchor	Bakery's ₹600 cake	A streaming plan alone; Flipkart's free-delivery rules
Value ceiling	₹900 for a birthday family	Delivery saved + video + music, for the benefits a member uses
Price chosen	₹700	₹1,499 a year (Lite ₹799, Shopping Edition ₹399)
Break-even for a cut	₹100 lost per cake × current sales ÷ profit per extra cake	Fee lost × current members ÷ value of a new member

3 · The framework. **Cost-plus** pricing starts from what it costs and adds a margin: it gives the floor. **Competitor** pricing looks at what alternatives charge: it gives the anchor customers compare with. **Value-based** pricing asks what the product is worth to the customer: it gives the ceiling. The price lives between anchor and ceiling (Figure 3.17). For a price change, add **price elasticity** (how much demand moves when price moves: if a 10% cut brings 5% more buyers, elasticity is 0.5) and the break-even: **new customers needed = revenue lost on existing customers ÷ profit per new customer.**

Using it, step by step (on this case):

1. Floor: work out what one customer costs to serve (free deliveries, content). Never price below it.
2. Anchor: find the price of the nearest alternative the customer would compare with (a streaming plan alone).
3. Ceiling: estimate the value to a *typical* customer, counting only the benefits they actually use.
4. Set the price between anchor and ceiling: nearer the ceiling if you are clearly better, nearer the anchor if you want share.
5. For a change: revenue lost on existing customers ÷ profit per new customer = new customers needed. Then test whether you can get them.

COMMON SLIP · adding up every benefit as if each customer uses all of them. That inflates the ceiling and makes almost any price look cheap.

Floor: cost per member

deliveries + content, illustrative ₹600–900

Ceiling: ₹4,400 if every benefit is used

Take B's sum; few members use all six



Figure 3.17 · Prime India on a price ladder. The real price sits just above the streaming anchor and well below the full-bundle value, because most members value only the one or two benefits they use. The cheaper plans sit near the floor. Prices: amazon.in, 2026. Anchor and value from Take B; floor and typical value illustrative.

4 · When it misleads. Value-based pricing overcounts when you sum benefits nobody uses together, as Take B did: a member who only wants fast delivery values Prime at about ₹600, not ₹4,400. And break-even maths ignores second effects: a cut can pull in bargain hunters who shop little (so each new member is worth less than \$80), or teach loyal members to wait for discounts.

5 · In the interview. “The price sits between what the alternative costs and what the product is worth to the user. For a cut, I’ll first work out how many new customers it needs just to break even, and whether that is realistic.”

E-COMMERCE & MARKETPLACES · GUESSTIMATES

Guesstimates: a number from nothing, out loud

“How many searches does Google answer each second?” Nobody expects you to know. The interviewer wants to watch you break an unknown number into pieces you can estimate, pick sensible values, and check the result. A wrong answer with a clear structure scores better than a right answer you can’t explain.

Four ideas that make every guesstimate easier

1. Top-down or bottom-up. **Top-down** starts from a big known number and narrows it: India’s 146 crore people → on-line → shopping on one app → orders a day. **Bottom-up** starts from one small unit and multiplies it out: one dark store’s pickers → orders an hour → stores in a city. Top-down is fast and suits “how many users or searches”. Bottom-up suits “how many stores, riders or servers”, where capacity is physical.

2. Demand side or supply side. Swiggy orders an hour can be counted from the **eaters** (households × how often they order) or from the **riders** (riders online × orders each carries an hour). They should agree. When they don’t, one of your assumptions is wrong, and finding which one is the best part of the answer.

3. Proxies. A **proxy** is a number you know that moves with the one you don’t. UPI’s 79 crore payments a day are a proxy for how often Indians pay digitally. A city’s number of two-wheelers is a proxy for potential bike-taxi riders. Name the proxy and say why it moves with the answer.

4. Round numbers and sanity checks. Use numbers that multiply easily: 146 crore becomes 150 crore, 86,400 seconds in a day become about 1 lakh (when precision matters, keep 86,400). Then check the answer *per person*: “13 billion searches a day is 3 per internet user”. If the per-person figure sounds silly, a step is wrong.

Number worth remembering	Round value	Source and date
India’s population	~146 crore	UN estimate, 2025
Indian households	~31 crore (≈ 4.7 people each)	Estimate from the population; the last census count was 2011
Active internet users in India	~96 crore	IAMAI-Kantar, 2025
People who shopped online in India	29–30 crore	Bain & Flipkart, 2025
UPI payments	24.5 bn a month ≈ 79 crore a day	NPCI, August 2026
People online worldwide	~600 crore (6 bn, 74%)	ITU, 2025
Seconds in a day · hours awake	86,400 ≈ 1 lakh · ~16	Arithmetic

The guesstimate kit

Start with a wedding caterer. A caterer in Lucknow is asked how much biryani to cook for a wedding. She doesn’t guess “100 kg”. She asks how many guests are invited (400) and what time dinner is served (late, so everyone eats). Most weddings see 80% turn up: 320 people. About 90% will take biryani: 290. At 250 g a plate that is 72 kg. She adds 10% because running out at a wedding is a disaster: about 80 kg. Then she checks: “last month, 350 guests ate 75 kg”. Same ballpark, so she cooks 80 kg. Every step of a good guesstimate is in that story.

1

Pin down the unit

What exactly are we counting, over what time, and where? Typical day or peak?

Caterer: plates of biryani, for one dinner, one wedding.

2

Pick the approach

Top-down or bottom-up; demand side or supply side. Say which, and why.

Caterer: bottom-up from guests (demand), not from pot sizes (supply).

3

Draw the tree

Write the answer as a product of 3–5 pieces before putting in any number.

Caterer: guests × turn-up × take biryani × grams a plate.

4

Round numbers, one reason each

Every number gets a one-line “because”. Round so the maths is easy aloud.

Caterer: 80% turn up, because that is what most weddings see.

5

Check it another way

A known figure, a per-person rate, or the other side (supply vs demand).

Caterer: last month 350 guests ate 75 kg.

6

Give a range, and what moves it

Name the one assumption that matters most, and what the answer would be if it were off.

Caterer: 70–90 kg; turn-up is the swing factor.

Rule of thumb: clarify in two or three questions, say your approach in one sentence, and write the tree down *before* the numbers. Interviewers check the tree first and the arithmetic second. The cards follow the steps taken by the strongest of the fourteen guess-timates in the IIM A and IIM B casebooks; the weaker ones skip card 5.

WHEN IT BREAKS

A tree with ten steps multiplies ten errors. If each step could be 20% off, the answer could be off by a factor of six. Keep the tree to 3–5 steps, and spend your time on the step that swings the answer most: for a store count, orders per household; for searches, searches per person.

IN THE INTERVIEW

“I’ll count this from the demand side, as users times searches per user, and then check it against what we know Google has said. The number I’m least sure of is searches per person, so I’ll show a range on that one.”

GUESSTIMATE 3.1 · GOOGLE SEARCHES PER SECOND

Guesstimate

Medium IIM B *Sigma PM Casebook 2026–27*, p. 141

Try it first. How many times did you search on Google yesterday, including voice and Lens? Is that more or less than a typical internet user worldwide? Build from there for five minutes.

THE QUESTION

Estimate the number of queries answered by Google per second.

A Surely. I have a few clarifying questions before we begin. Is there any particular geography that I should focus on? And also any particular mode of search like via the mobile or desktop?

Q You can consider the global population, and both modes of search are acceptable.

A Thank you. I’ll go with a top-down approach starting with the global population, incorporate for certain factors like internet penetration, the market share for google, and frequency of search for different types of users. Does that sound fair?

Q Sure, can you assign some numbers to these factors?

A The global population is close to 8 Bn people but as far as I’m aware, Google search is blocked in China, so that takes away a big chunk of 1.4 Bn away leaving 6.6 Bn people. Internet penetration globally is ~ 60% and that gives the total number of internet users as 6.6 Bn x 60% = 4 Bn (approx.) I know there are a lot of other search engines available, but Google holds a big chunk of that market. Is it fair to assume that Google’s market share in search is 90%?

Q Yes, you can proceed with that.

A Thank you. So, now I have the total number of users of Google search as 4 Bn x 90% = 3.6 Bn. Next, I would like to incorporate some user personas because different people will have differing usage. For instance, certain segments like students at a higher level of education, or professionals in fields like research, coding, etc. would use Google search extensively for their day-to-day tasks. I can segment the population into the following 3 groups -

Description	Split	Number	Queries / day	Total number of queries / day
Power User	15%	0.54 Bn	10	5.4 Bn
Casual User	70%	2.52 Bn	5	12.6 Bn
Occasional User	15%	0.54 Bn	2	1.08 Bn
Total		3.6 Bn		19.08 Bn

A From this I can get the total number of queries being searched on Google as ~19 Bn. To get the number of queries per second we can simply divide this by 24x60x60. Therefore, number of queries answered by Google per second = 19 Bn / (24x60x60) = 220K (approx.)

Q You’ve used a good structured approach, but can you think of anything that you may have missed?

- A** Recently we've seen the rise of many AI powered search engines like ChatGPT and Perplexity.AI that people are now shifting to for getting answers to many of their queries. So, the number of queries being answered by Google would have taken a slight hit.
- Q** That's right, we have actually observed an average drop of 10% per user for our search engine in recent times because of many more AI tools in the market.
- A** Got it, so the total number of queries answered would come down to $220K \times 90\% = 200K$ (approx.)
- Q** Thank you, we can close the case.

WHAT WORKED · WHAT WAS MISSING

- ✓ **Clarified geography and device**, then stated the approach before any number.
- ✓ **Removed China**, where Google is blocked: a sharp, specific step.
- ✓ **Split users by how much they search** (power, casual, occasional) instead of one average.
- ✓ **Adjusted for AI chat tools** when prompted, and redid the maths.
- ✗ **Never checked against a known figure.** Google said in 2025 that it handles "more than 5 trillion searches a year", about 158,000 a second.
- ✗ **No per-person sanity check.** 19 billion ÷ 3.6 billion users is 5.3 searches a day each. Plausible, but it should be said out loud.
- ✗ **Average or peak?** For servers, the peak hour matters, roughly 1.5–2× the average.
- ✗ **Undercounted people online:** 60% of the non-China world, when ITU puts worldwide use at 74% in 2025.

Guesstimate kit check: 1 Unit ✓ (geography, device) · 2 Approach ✓ top-down · 3 Tree ✓ · 4 Numbers with reasons ✓ · 5 Check another way ✗ · 6 Range ✗. Skipping card 5 cost the most: a one-line check against Google's own figure would have shown the answer was about 25% high, and why.

CHECK · "INTERNET PENETRATION GLOBALLY IS ~60%" DIRECTIONALLY RIGHT

ITU estimates about 6 billion people online in 2025, 74% of the world. Take away about 1.1 billion online in China and roughly 4.9 billion remain, against the candidate's 4 billion. The error happened to cancel out a high searches-per-person figure, so the answer landed close. Two errors that cancel are luck, not method.

STRONGER ANSWER



Figure 3.18 · The six kit cards, applied to Google searches. The check against Google's published figure (card 5) is what lets you state the answer with confidence. *Author's summary.*

- A 1 · UNIT**
I'll count every query Google answers, typed, spoken or through Lens, worldwide, averaged over a typical day. Then I'll say what the busiest hour looks like, because servers are built for the peak.
Why this move: "per second" can mean average or peak; saying which avoids a mismatch with what the interviewer had in mind.
- A 2 · APPROACH**
Top-down, from the demand side: the number of people who use Google, × searches each person makes a day, ÷ the seconds in a day.
Why this move: people and habits are easier to estimate than Google's servers, and a published figure exists to check against.
- A 3–4 · TREE AND NUMBERS**
About 6 billion people are online (ITU, 2025). About 1.1 billion of them are in China, where Google is blocked: $6 - 1.1 = 4.9$ billion. Google has about 90% of search: $4.9 \times 90\% \approx 4.4$ billion searchers. On average each searches about 3 times a day: heavy users do 10 or more, many do one or none. $4.4 \text{ billion} \times 3 = 13 \text{ billion searches a day}$. $13 \text{ billion} \div 86,400 \text{ seconds} \approx 150,000 \text{ a second}$ (Figure 3.19).
Why this move: three multiplications, each with a one-line reason, are easy for the interviewer to follow and to challenge.
- A 5 · CHECK**
Google said in 2025 that it handles more than 5 trillion searches a year. $5 \text{ trillion} \div 365 \approx 13.7 \text{ billion a day}$; $\div 86,400 \approx 158,000 \text{ a second}$. My 150,000 is within 5% of that.
Why this move: a second route to the answer turns a guess into an estimate you can defend.
- A 6 · RANGE**
The number I'm least sure of is searches per person. At 2 a day the answer is about 100,000 a second; at 5 a day, about 250,000. At the busiest hour traffic runs roughly 1.5–2× the average, so Google must be built for about 250,000–300,000 a second. AI chat tools push the per-person number down; AI Overviews and Lens push it up.
Why this move: a range with its main driver shows judgement, and tells the listener what to watch.

HOW THE ANSWER HANGS TOGETHER

The unit (average, all query types, worldwide) set up a simple demand-side tree. Each number has a reason, the check against Google's own figure confirms the scale, and the range names the one input that moves it.

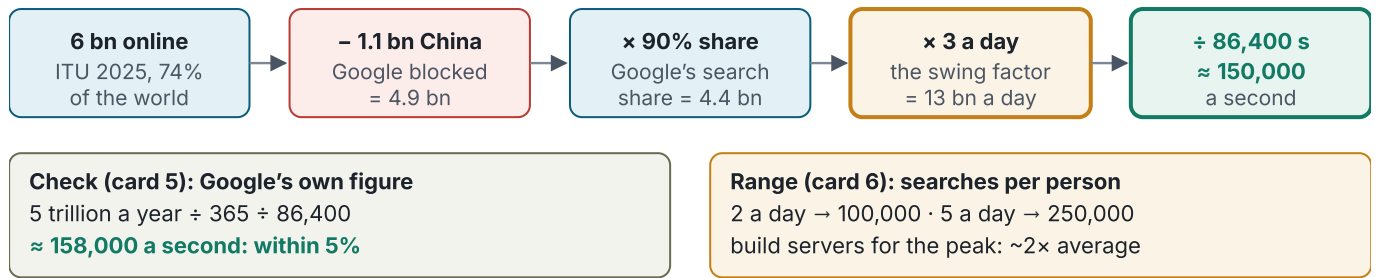


Figure 3.19 · Google searches per second, as an equation tree. Four steps, one swing factor (orange), and a check against a published number. The amber box is where to spend your time. ITU Facts and Figures 2025; Google's "5 trillion searches a year" (announced March 2025); search share rounded from StatCounter.

THE REUSABLE TRICK

Per second = per day ÷ 86,400, which is about 1 lakh. So "a billion a day" is about 11,500 a second, and "10 crore a day" (UPI is near 79 crore) is about 1,150 a second. Remember one pair and you can convert any daily number to a per-second rate in your head.

GUESSTIMATE 3.2 · DARK STORES FOR 10-MINUTE AHMEDABAD Guesstimate Hard IIM A Product Mindset 2026–27, pp. 240–241

Try it first. A rider on a scooter has 7 minutes to reach any home from the store. How far can they go in Ahmedabad's evening traffic? Turn that into an area per store before reading on.

THE QUESTION

How many dark stores would a quick-commerce player need in Ahmedabad to guarantee 10- minute delivery across the city?

- A** Three clarifications first. Is this the city, or the full district?
- Q** City plus immediate suburbs - the area a player would realistically serve.
- A** A single player, or everyone combined?
- Q** A single dominant player, covering the area on its own.
- A** And does the 10 minutes include pick and pack, or is it pure travel?
- Q** Full order-to-doorstep window.
- A** Then I will solve this twice. One limit is coverage - how much ground a single store can reach in 10 minutes. The other is throughput - how many orders that store can handle in its busiest hour. I need to satisfy both, so the answer is the larger number. May I start with coverage?
- Q** Please go ahead.
- A** Path 1 - Coverage
Of the 10 minutes, 3 go to pick, pack and rider handoff. That leaves 7 minutes of travel. For serviceable area I will use 250 sq km.
- Q** Ahmedabad municipal limits are considerably larger than that. Justify the number.
- A** Fair. Quick commerce only works where order density supports a store. That means the denser residential and commercial belt - the western side around SG Highway, Satellite, Bodakdev and Vastrapur, plus the denser central and eastern areas. Not the industrial fringe. It is also my most load-bearing assumption here, and the coverage answer scales linearly with it. Full municipal coverage would roughly double this number
- Q** Accepted. Give me the calculation.

Coverage maths	Value
Travel time inside the promise	7 min
Two-wheeler speed in mixed traffic	20 km/h → 2.3 km road distance
÷ circuitry factor 1.3 (roads bend)	1.8 km straight-line radius
Area per store (πr^2)	≈ 10 sq km
250 ÷ 10, plus ~10% because circles do not tile	≈ 28 stores

The coverage table is printed beside "Path 2" in the casebook; it is moved up to follow the coverage path it belongs to.

A Path 2 - Throughput

Now store capacity, built from the floor up. I will assume 8 pickers on the peak shift, at 15 orders per hour each. That is 120 orders in the peak hour. The obvious next step is wrong. Multiplying by a 16-hour day gives 1,920 orders. But that assumes flat demand, and grocery delivery has a sharp evening rush. The busiest hour carries roughly 15% of the day.

Q Correct - most candidates multiply by opening hours and stop there. Go on

A Staffing does not flex minute to minute, so the peak hour is the ceiling. Effective daily capacity = $120 \div 0.15 \approx 800$ orders per day. Now the demand side.

Demand maths	Value
Ahmedabad urban population $\div$ household size 4	~23 lakh households
$\times$ 17% active quick-commerce penetration	~3.9 lakh active households
$\times$ 0.3 orders per active household per day	~1.2 lakh orders/day, all players
$\times$ 35% share for a category leader	~42,000 orders/day for our operator
$\div$ 800 orders per store per day	$\approx$ 53 stores

Q Hold on. 0.3 orders per household per day is 9 a month. Defend that for Ahmedabad.

A Fair challenge. "Active household" is a wide net - anyone who has ordered recently, including someone topping up twice a month. Heavy users in SG Highway or Satellite order three or four times a week. But they are a minority. Most of Ahmedabad still does the main grocery run at a kirana. Blended across the whole active base, 0.3 is defensible. It is also the number I am least confident in, so let me show you what it costs me if I am wrong

Q Please do

Orders per active household/day	Operator orders/day	Stores needed
0.2 (6 a month)	~27,000	$\approx$ 34
0.3 (9 a month) - base case	~42,000	$\approx$ 53
0.4 (12 a month)	~55,000	$\approx$ 68

A A one-third error in that number swings the store count by about 35%. Penetration, frequency and market share are all multiplicative, so they behave the same way. Store capacity is the only input I built from something observable, so it is the one I would defend hardest.

Q Fine. Keep going with 0.3.

Reconciling

Q Your two numbers disagree. 28 against 53. Which one is wrong?

A Neither. They measure different constraints, and the gap is the most useful thing here. I have to satisfy both, so the answer is the larger. Demand binds, not geometry. That has a real consequence. The operator is not adding stores to reach farther neighbourhoods; 28 would do that. It is adding 25 more to stop stores overflowing at peak. So those extra stores cluster in the dense pockets rather than spreading evenly across the map.

Q Sanity-check 53 for me before we go further.

A 53 stores across 250 sq km is one every 2.2 km, and one per 1.75 lakh residents. Each does about 790 orders a day, which matches the 800 capacity I built. The two paths reconcile.

Recommendation

Q So would you build 53?

A No. 53 is the theoretical minimum, and I would not build to it. At 53 stores the network handles 42,000 orders against 42,400 of capacity. That is 99% utilisation with no headroom. One rainy evening or one festival weekend and the peak hour breaks. Sizing for 80% peak utilisation instead gives $42,000 \div 640$, or about 66 stores. So I would plan for roughly 65, and phase them. Open the 28 that give geometric coverage first. Prove store-level economics there. Then add density in the high-order pockets as demand appears, rather than funding 25 half-empty stores up front.

WHAT WORKED · WHAT WAS MISSING

- ✓ **Three clarifications that each changed the maths:** city or district, one player or all, and whether the 10 minutes include picking.
- ✓ **Solved it twice, for two different limits** (reach and peak capacity), and took the larger.
- ✓ **Used the peak hour, not a flat day:** 120 orders an hour is 800 a day, not 1,920.
- ✓ **Showed a sensitivity table** for the weakest number and said which input it trusted most.
- ✗ **No check against real store counts.** Mapped data for mid-2025 shows Blinkit with 44 stores in Ahmedabad and all players with about 122.
- ✗ **Used daytime speed for the evening peak.** At 15 km/h instead of 20, the coverage count rises to about 53, the same as the demand count.
- ✗ **Circles need about 20% overlap to cover a map, not 10%.** Small, but it moves coverage from 28 to about 30 stores.

✓ **Refused to build to 99% utilisation**, sized for 80%, and phased the build.

Guesstimate kit check: 1 Unit ✓ · 2 Approach ✓ **supply and demand** · 3 Tree ✓ · 4 Numbers with reasons ✓ · 5 Check another way ✓ **(the two paths)**, but not against real counts ✗ · 6 Range ✓. One of the strongest answers in either casebook; what it skipped is a real-world check that would have confirmed it.

CHECK · “DEMAND BINDS, NOT GEOMETRY” **DIRECTIONALLY RIGHT**

True at daytime speeds. But the evening rush is exactly when demand peaks *and* scooters slow down. At 15 km/h, 7 minutes cover 1.75 km of road, or 1.35 km in a straight line; each store reaches about 5.7 sq km, and 250 sq km needs about 53 stores. At the peak, both limits bind at once. Real networks confirm the scale: Blinkit, the leader, had 44 mapped stores in the city in mid-2025.

STRONGER ANSWER



Figure 3.20 · The six kit cards, applied to dark stores. The candidate already did cards 1–4 well; the stronger version adds the peak-speed check and a real-world count. *Author’s summary.*

A 1–2 · UNIT AND APPROACH

Stores for one leading player, covering the dense 250 sq km of the city, with order-to-door in 10 minutes. Two things limit a store: how far its riders can reach in time, and how many orders its pickers can pack in the busiest hour. I’ll work out both and take the larger number.

Why this move: a store count must satisfy every limit at once, so the answer is the tighter of the two.

A 3–4 · REACH, AT THE PEAK

Picking and packing take about 3 minutes, which leaves 7 minutes to ride. In evening traffic a scooter averages about 15 km/h: 7 minutes $\times$ 15 km/h $\approx$ 1.75 km of road. Roads bend, so the straight-line reach is about $1.75 \div 1.3 \approx 1.35$ km. Each store’s circle is $\pi \times 1.35^2 \approx 5.7$ sq km, but circles must overlap to leave no gaps, so each store really covers about 83% of that: ≈ 4.7 sq km. $250 \div 4.7 \approx$ **53 stores** (Figure 3.21).

Why this move: the slowest traffic and the most orders arrive in the same hour, so reach must be tested at the peak, not at noon.

A 3–4 · CAPACITY

At the peak, 8 pickers $\times$ 15 orders an hour = 120 orders an hour per store. The busiest hour carries about 15% of the day, so a store handles about $120 \div 0.15 = 800$ orders a day. Demand: 23 lakh households $\times$ 17% who order $\times$ 0.3 orders a day $\times$ 35% share $\approx$ 42,000 orders a day. $42,000 \div 800 \approx$ **53 stores**.

Why this move: the peak hour, not the whole day, is what a fixed team of pickers can handle.

A 5 · CHECK

Mapped data from mid-2025 shows Blinkit with 44 stores in Ahmedabad and all players with about 122. So the leader runs about 36% of the city’s stores, close to my 35% share. My 53–65 for a player at full maturity sits sensibly above today’s 44.

Why this move: a real count confirms the scale and catches any assumption that is badly off.

A 6 · RANGE

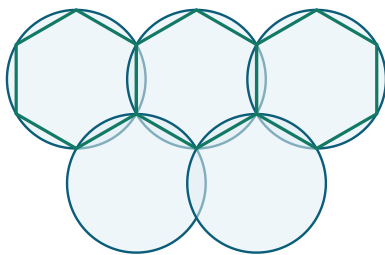
I’d plan about 65 stores, so each store runs at about 80% of its peak capacity rather than 99%. The range is 45–70, and orders per household is the number that swings it most. I’d open about 30 for coverage first, then add stores in the busiest neighbourhoods as orders grow.

Why this move: headroom absorbs rainy evenings and festivals; phasing avoids paying for half-empty stores up front.

HOW THE ANSWER HANGS TOGETHER

Both limits were worked out at the same moment, the evening peak, and both land near 53. The real count (Blinkit’s 44) confirms the scale, headroom lifts the plan to about 65, and the range names the one input to watch.

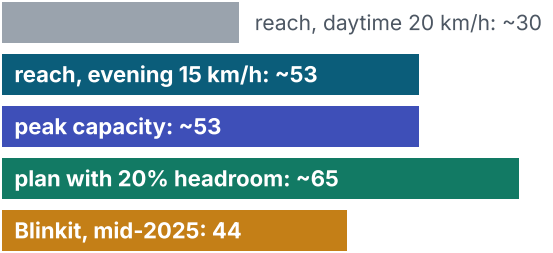
Why a store covers less than its circle



circles must overlap to leave no gaps; each store really "owns" the green hexagon, ~83% of its circle

radius 1.35 km at 15 km/h
→ circle 5.7 sq km
→ hexagon 4.7 sq km

Two limits, one answer



bar length = number of stores (1 store = 3.9 units)

Figure 3.21 · How many dark stores does a 10-minute promise need? Left: circles overlap, so each store effectively covers a hexagon about 83% of its circle. Right: at evening speeds the reach limit and the capacity limit meet at about 53 stores; the leader's real count in 2025 sits just below. Case numbers (IIM A p. 240); store counts from quickcommerce.com (third-party mapping, mid-2025, approximate).

THE REUSABLE TRICK

For anything built from physical capacity (stores, riders, servers, counters), size for the **peak hour**, not the day, and leave about 20% headroom. Then check the two limits, reach and throughput, at the same moment: the busiest hour is usually also the slowest.

Two guesstimates, one kit

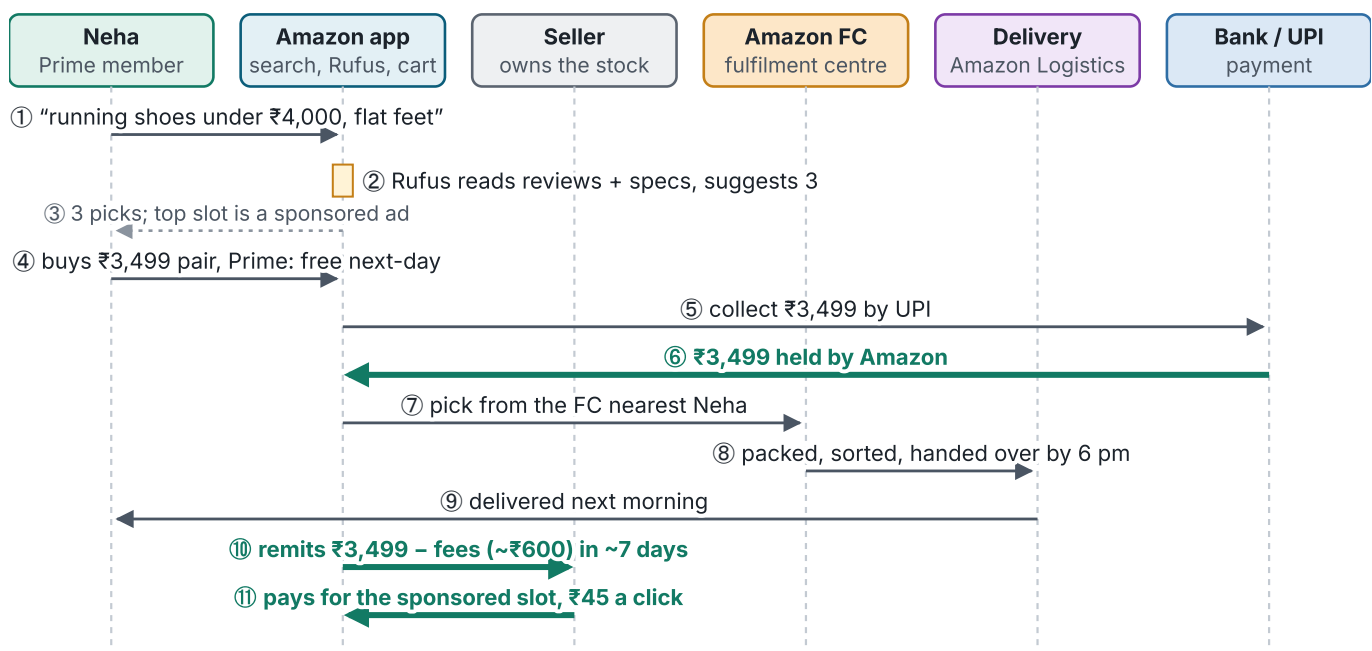
Card	Google searches a second (demand side)	Dark stores for Ahmedabad (supply side)
1 · Unit	All queries answered, worldwide, average second; peak stated separately	Stores for one leading player, 10 minutes door to door, the dense 250 sq km
2 · Approach	Top-down from people: users × searches each	Bottom-up from one store: its reach and its pickers
3–4 · Swing factor	Searches per person a day (2 to 5)	Orders per active household a day (0.2 to 0.4)
5 · Check	Google's "5 trillion a year" ≈ 158,000 a second	Blinkit's 44 mapped stores (mid-2025)
6 · Answer	~150,000 a second (100,000–250,000); build for ~300,000 at peak	~53 stores at the limit; plan ~65 (45–70)

Pick the approach by what is **physical**. Searches, users and payments are counted from people (demand). Stores, riders, servers and counters are limited by space, time and staff (supply). When both sides can be estimated, do both; the gap between them is where the interesting discussion starts.

Amazon: the store that is really a delivery network and an ad business

Walkthrough (illustrative persona). Neha, 31, a Prime member in Bengaluru, needs running shoes for flat feet. She types the question into the Amazon app's AI assistant (launched as Rufus, now merged with Alexa+ into one shopping assistant). It reads reviews and product specs and suggests three pairs; the first result is a sponsored listing. She buys a ₹3,499 pair with free next-day delivery because she's a Prime member. The shoes come from a fulfilment centre (FC: Amazon's warehouse, where sellers store stock) 18 km away. She never thinks about the seller, who is paid about a week later (Figure 3.22).

Feature	User problem	Why it's designed this way	Metric it moves
Prime	"I don't want to pay for delivery every time"	One yearly fee (₹1,499) makes Amazon the default shop; the video and music in the bundle keep members renewing even in months they buy little	Orders per member; renewal rate
AI shopping assistant	"Which of these 200 shoes is right for me?"	Answers questions in plain words from reviews and specs, then links to products; it shortens the "compare" step of the funnel	Search-to-purchase conversion
Sponsored Products	Sellers: "nobody sees my listing on page 5"	Sellers bid per click for a slot in search results; the auction weighs relevance so ads don't push shoppers away	Ad revenue; conversion on ad clicks
Fulfilled by Amazon	Sellers can't promise next-day delivery themselves	Sellers store stock in Amazon's FCs and get the Prime badge; Amazon controls speed and quality	Share of orders delivered next day
Amazon Now	"I need it in the next hour"	Small city stores deliver everyday items in minutes, so Amazon keeps the frequent purchases quick commerce was taking	Orders per customer per month



Amazon earns four ways on one order: seller fees, fulfilment, the ad click, and the Prime fee that made Neha buy here.

Figure 3.22 · Behind Neha's shoe order. One ₹3,499 order earns Amazon in four places: the seller's referral and fulfilment fees (⑩), the ad click (⑪), and the Prime fee that made Neha buy here rather than elsewhere. *Simplified from Amazon seller documentation; fee and bid amounts illustrative.*

Core loop: the flywheel

Jeff Bezos sketched Amazon's strategy on a napkin in 2001 as a loop with no start and no end (Figure 3.23). More selection gives shoppers a better experience; more shoppers attract more sellers; more orders spread the cost of warehouses and trucks, which allows lower prices, which brings more shoppers. Prime adds a habit to the loop: once you have paid the fee, you shop at Amazon first. And in the last decade ads were added on top: more shoppers make every sponsored slot more valuable, and that ad money pays for faster delivery.

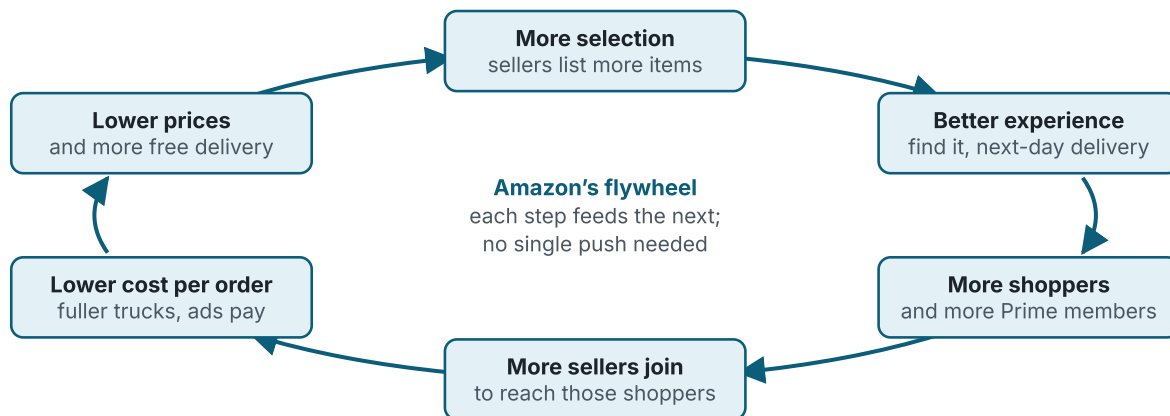


Figure 3.23 · Amazon's flywheel. Push any box and the wheel turns faster; for example, fuller trucks lower the cost per order, which funds lower prices. Competitors copy one box at a time; the advantage is having all six. *Adapted from Amazon's published flywheel (2001 onwards).*

Business model, with numbers

Amazon doesn't report India on its own. Worldwide, in April–June 2026, it had revenue of **\$200.6 bn** (+20%). Two lines matter most for understanding its shop: **advertising, \$19.8 bn** (+26%), almost all profit, and **AWS, \$42.2 bn** (+37%), its cloud business, which earns most of the company's operating profit. In India, the store earns from seller fees (referral, fulfilment, closing fees), ads, Prime fees and payments. From 16 March 2026 it removed referral fees on items under ₹1,000 across 1,800+ categories, matching Flipkart and Meesho; sellers now pay mainly through fulfilment fees and ads.

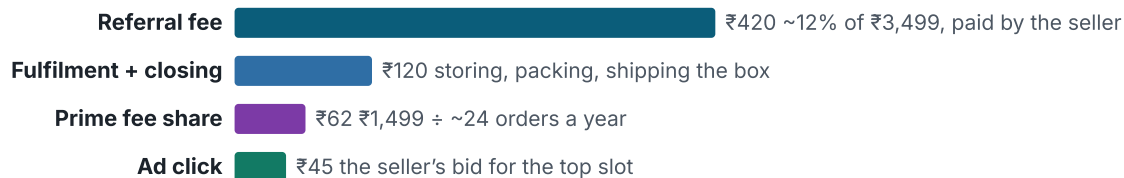


Figure 3.24 · Four ways Amazon earns on Neha's ₹3,499 shoes. The referral fee is the biggest single item, but ads and the Prime fee are the ones that grow fastest and cost least to deliver. *Illustrative: referral ~12% of the price; fulfilment and closing fees typical for a shoe box; one ad click; ₹1,499 Prime fee spread over ~24 orders a year.*

Moat, in plain words. The delivery network: a nationwide network of fulfilment centres, sort centres and delivery stations took a decade and billions of dollars to build, and they let Amazon promise next-day delivery. The Prime habit: members check Amazon first. And the ad data: Amazon knows what people actually *buy*, not just what they look at, which makes its ads worth more to brands.

NORTH STAR	INPUTS	GUARDRAILS
Weekly purchases by repeat customers.	In-stock selection with next-day delivery · search-to-purchase conversion · Prime sign-ups and renewals.	Sponsored share of the first screen vs conversion · return rate · seller fees as % of price.

2025–26 moves. Zero referral fee under ₹1,000 (March 2026). Amazon Now's 10-minute delivery expanding across Indian cities. Rufus and Alexa+ merged into one AI shopping assistant; Amazon said active users nearly doubled and interactions rose more than 5× in a year (Q2 2026).

STRENGTHS

- ✓ Fastest reliable delivery; the Prime habit
- ✓ Ads and AWS pay for price cuts and speed
- ✓ Trusted for electronics and branded goods

WEAKNESSES

- ✗ Weaker than Flipkart in smaller towns and in the big October sale
- ✗ Too many ads on the first screen can hurt conversion
- ✗ FDI rules limit how much it can control stock and prices

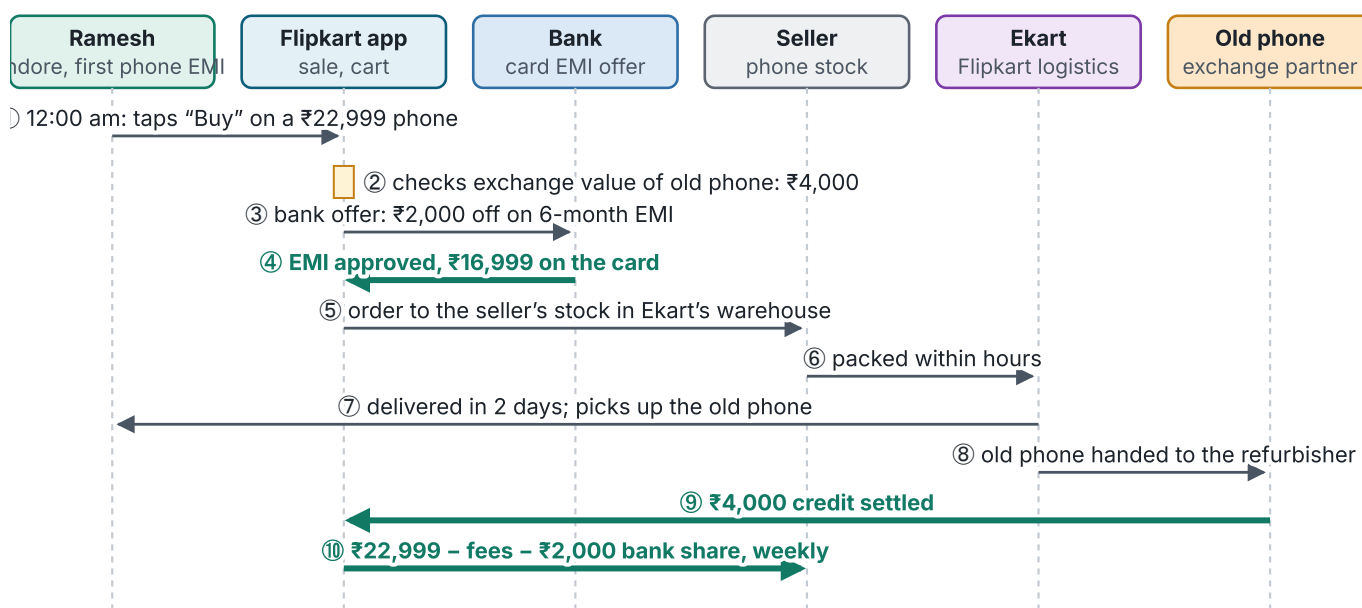
What I'd build next

1. Tier-2 shoppers don't trust fit or quality for clothes → "Kept by people like you": show the share of buyers in your size who kept the item, next to the price → return rate on fashion; conversion.
2. The AI assistant answers questions but can't follow up after delivery → the assistant asks "did they fit?" three days later and offers an exchange in one tap → exchanges instead of refunds; repeat purchases.
3. Prime members in small towns see little value in video → promote the ₹399 Shopping Edition to them, with local-language delivery tracking → paid members outside metros; orders per member.

Flipkart: India's biggest sale, and the fight for every day in between

Walkthrough (illustrative persona). Ramesh, 38, runs a hardware shop in Indore. His phone is four years old. At midnight on the first day of Big Billion Days, the ₹22,999 phone he has watched for weeks becomes affordable: the app values his old phone at ₹4,000, his bank card takes ₹2,000 off if he pays in six monthly instalments (EMI), and he pays ₹16,999 spread over six months. Two days later an Ekart rider hands over the new phone and takes the old one away (Figure 3.25).

Feature	User problem	Why it's designed this way	Metric it moves
Big Billion Days	"I wait all year for the best price on a phone or TV"	A few days where brands, banks and Flipkart all fund discounts; early access for members creates urgency	Sale-week GMV; new buyers won
EMI and bank offers	"₹23,000 at once is too much"	Banks pay part of the discount to win card spending; EMI turns a big price into a monthly one	Conversion on phones and appliances
Exchange offer	"What do I do with my old phone?"	A refurbishing partner values and collects the old phone at delivery, so the upgrade feels cheaper	Upgrade purchases; order value
Shopsy	"I want very low prices"	A separate value app, so low-price sellers don't cheapen the main brand	Buyers won from Meesho's segment
Minutes	"I need it now"	~1,000 dark stores in ~130 cities (mid-2026) keep everyday buying inside Flipkart	Orders per customer per month



A sale-day phone is a bundle of four deals: the bank's EMI discount, the exchange, the seller's price and Flipkart's delivery.

Figure 3.25 · Behind Ramesh's sale-night phone. The ₹6,000 that made the phone affordable came from two outside parties: the bank (3–4) and the refurbisher who buys the old phone (8–9). Flipkart's skill is assembling these deals for millions of orders in one night. *Simplified; amounts illustrative, typical of 2025 sale offers.*

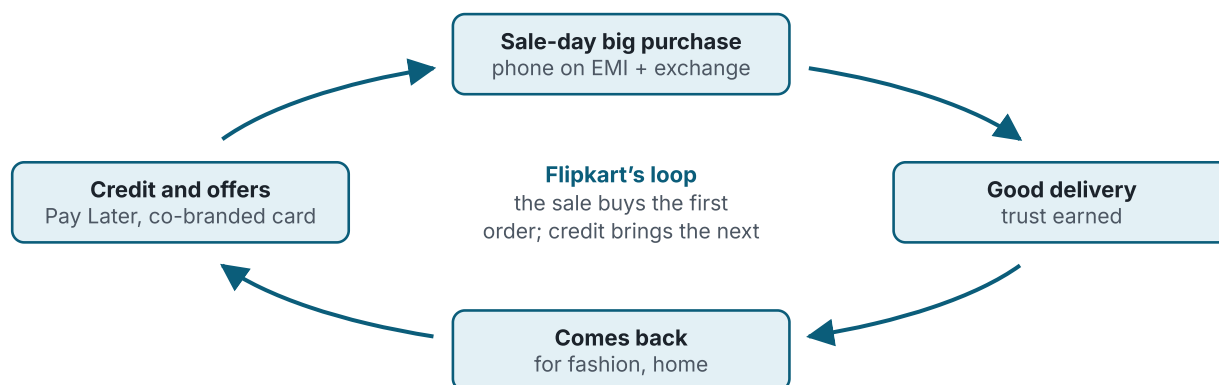


Figure 3.26 · Flipkart's loop. The sale wins the first big order; credit and offers bring the next ones. The weak link is the step in between: coming back on ordinary days. *Author's summary.*

KEY IDEA · WHY THE SALE IS NOT ENOUGH

A phone is bought every two or three years. A shopper who comes only for the October sale is worth one big order and then disappears to Blinkit for chargers, to Meesho for sarees and to Amazon for everything else. Flipkart's bets since 2024 (Minutes, Shopsy, super.money, zero commission under ₹1,000) all aim at the same thing: **getting the sale-day customer to come back in March.**

Business model, with numbers. Flipkart earns from seller fees, logistics fees (Ekart), ads and its fintech partners. Its marketplace company reported FY25 revenue of **₹20,493 Cr** (+14%) and a loss of **₹1,494 Cr**. In November 2025 it dropped commission on goods under ₹1,000; it said sellers' costs in that range would fall by about 30%. It has moved its holding company back to India, a step before a stock-market listing that has been pushed back more than once. Walmart has owned most of it since 2018.

NORTH STAR	INPUTS	GUARDRAILS
Buyers who order in at least three different months a year.	Purchases outside sale weeks · share of orders from non-phone categories · Minutes orders · EMI and Pay Later users.	Return and RTO rate · seller satisfaction · site uptime on sale night.

Moat, in plain words. Phones and large appliances: exclusive launches, the exchange network and bank tie-ups are hard to copy. Ekart reaches towns rivals find costly to serve. And Walmart's money lets it fight Meesho on price and Blinkit on speed at the same time. The risk is that it fights on too many fronts.

STRENGTHS

- ✓ The biggest sale event; strongest in phones
- ✓ Reach in Tier-2 and Tier-3 India
- ✓ Myntra, a profitable sister company

WEAKNESSES

- ✗ Still loss-making; listing delayed
- ✗ Weak everyday frequency
- ✗ Squeezed by Meesho on price and quick commerce on speed

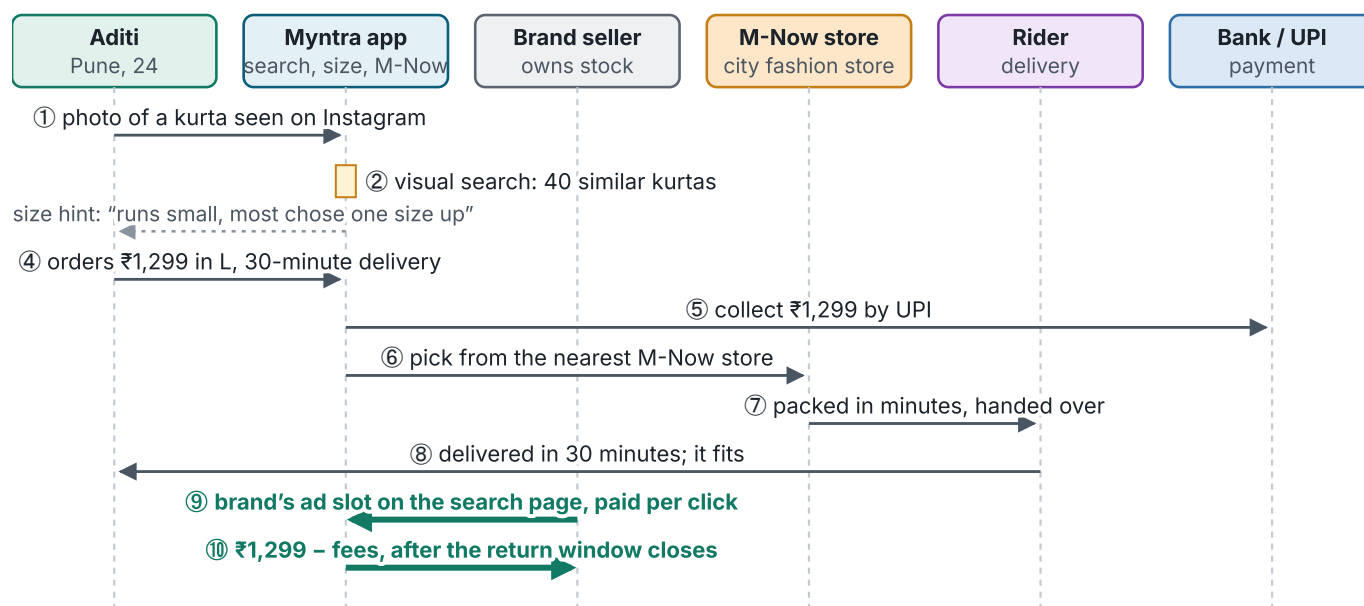
What I'd build next

1. Phone buyers vanish for two years → a phone care plan: protection, repair pickup and a guaranteed buy-back value, with an upgrade reminder → repeat phone purchases; care-plan attach rate.
2. Sale-day buyers don't know Minutes exists → a free first Minutes order (charger, case, screen guard) inside the phone's delivery message → share of sale-day buyers who place a second order within 30 days.

Myntra: one category, done deeply, turned into profit

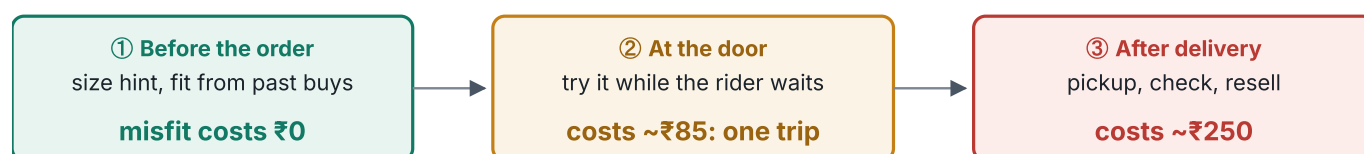
Walkthrough (illustrative persona). Aditi, 24, works in Pune. She sees a kurta on an influencer's Instagram post, screenshots it, and uploads the photo to Myntra's visual search. Forty similar kurtas appear. On the one she likes, a small line says "runs small: most people chose one size up", so she orders an L instead of her usual M. It is in stock at an M-Now store nearby, so it arrives in 30 minutes. It fits (Figure 3.27).

Feature	User problem	Why it's designed this way	Metric it moves
Visual search	"I saw it on Instagram but can't describe it"	A photo replaces words; computer vision finds items with a similar cut, colour and pattern	Searches that end in a product view
Size and fit hints	"Will it fit?"	Returns data per product turns into a one-line hint before the order, the cheapest point to stop a misfit	Return rate for size reasons
M-Now	"I need an outfit for tonight"	Fashion and beauty in about 30 minutes from city stores; launched December 2024	Orders per customer; share of urgent occasions won
MyFashionGPT	"What should I wear to a beach wedding?"	Launched 2023: type a need in plain words, get a set of products instead of a keyword search	Search success on vague queries
Brand ads	Brands: "we need to be seen during the sale"	Sponsored slots and brand pages; fashion shoppers browse, so there are many slots to sell	Ad revenue (₹915 Cr in FY25, +28%)



The size hint is the cheapest return-cutting tool Myntra has: it acts before the order, not after.

Figure 3.27 · Behind Aditi's kurta. Two steps do the heavy lifting: the photo search (2) that finds the product without words, and the size hint (3) that makes the kept order more likely. The brand pays for the slot (9) whether or not Aditi keeps the kurta. *Simplified from Myntra's public feature descriptions; amounts illustrative.*



The earlier a misfit is caught, the cheaper it is: move effort to the left.

Figure 3.28 · Three places to catch a kurta that won't fit. Every rupee spent improving size hints saves more than the same rupee spent on faster return pickups. *Costs illustrative, from the return waterfall in the primer.*

Business model, with numbers. Myntra earned **₹6,043 Cr** in FY25 (+18%) and a profit of **₹548 Cr**, up from ₹31 Cr, helped by about ₹138 Cr of deferred tax credit. Half its revenue is logistics fees charged to sellers, a third is commission, and ads bring ₹915 Cr. The lesson for PMs: a focused vertical with high-margin ads and loyal 18–35-year-olds can make money where the everything-stores can't. More than 70% of its new customers in 2025 came from outside the metros.

NORTH STAR Kept orders per active customer per quarter.	INPUTS Search-to-cart rate · size-hint coverage of catalogue · M-Now orders · new customers outside metros.	GUARDRAILS Return rate by reason · ad load per search page · brand partner satisfaction.
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Moat, in plain words. Long relationships with fashion brands, years of fit and return data per product, and a young audience that browses often, which makes its ad slots valuable. **2025–26 moves:** M-Now scaling across cities; more AI-led discovery; growth outside metros.

STRENGTHS

- ✓ Profitable, unlike most rivals
- ✓ Deep fashion data and brand ties
- ✓ Ads grow fast at low cost

WEAKNESSES

- ✗ High return rates are built into the category
- ✗ Meesho and Shopsy win the price-conscious shopper
- ✗ Depends on a few big sale events

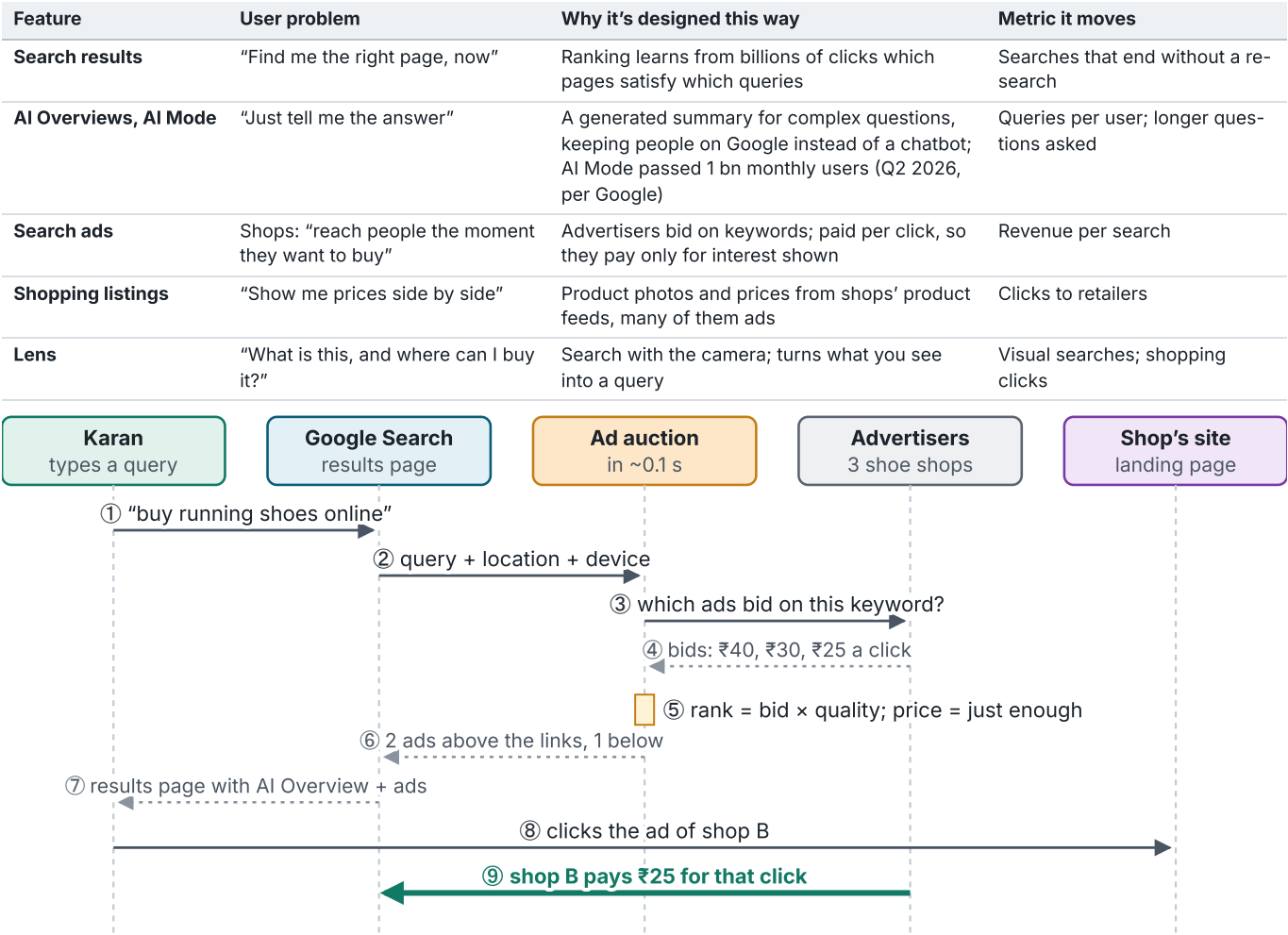
What I'd build next

1. Customers order two sizes and return one → “your size here”: fit advice from the customer's own kept orders, shown when a second size is added → bracketed orders; size-related returns.
2. M-Now orders are urgent and misfits hurt most → try-at-the-door on M-Now orders only, where the rider is nearby anyway → kept rate on M-Now; cost per M-Now order.

Google Search: people say what they want, and advertisers pay to answer

Walkthrough (illustrative persona). Karan, 27, in Hyderabad, types “buy running shoes online” into Chrome. Within a quarter of a second he sees an AI Overview summarising what to look for, two ads above the ordinary

results, and a row of shopping listings. He clicks the second ad, from a shoe shop, and lands on its page. In that quarter-second an auction ran among every advertiser bidding on his words, and the shop agreed to pay Google only because he clicked (Figure 3.29).



Google is paid only when Karan clicks, so it wants the ad he is most likely to click, not only the highest bid.

Figure 3.29 · Behind one search with ads. The auction (③–⑤) runs for every search that has ads, in a fraction of a second. Google earns nothing unless Karan clicks (⑧–⑨), which is why it cares about which ad he will click, not only who bids most. *Simplified from Google Ads documentation; amounts illustrative.*

How the search ad auction works

Google doesn't simply sell the top slot to the highest bidder. Each ad gets a score: **Ad Rank** $\approx$ **bid** $\times$ **quality**, where quality is Google's estimate of how likely people are to click and how useful the landing page is. The winner pays only enough to keep its place above the next ad. So a relevant ad can win with a lower bid, and pay less (Figure 3.30). Everyone gains: users see more useful ads, good advertisers pay less, and Google earns more clicks overall.

Three shoe shops bid for "buy running shoes online" (illustrative)

Shop A	bid ₹40 × quality 5		rank 200
Shop B	bid ₹30 × quality 8		rank 240: top slot
Shop C	bid ₹25 × quality 6		rank 150

B wins with the lower bid, and pays just enough to beat A: $200 \div 8 = ₹25$ a click (not its ₹30 bid).
A comes second and pays enough to beat C: $150 \div 5 = ₹30$. The best ad gets the best slot at the lowest price.

Figure 3.30 · Why a better ad can pay less. Multiplying the bid by quality rewards ads people actually click. The real auction adds reserve prices, ad formats and other signals, but this is its core. *Simplified from Google Ads' published Ad Rank explanation; numbers illustrative.*

Business model, with numbers. In April–June 2026 Alphabet earned **\$119.8 bn** (+24%). **Search and other: \$63.3 bn** (+17%), still growing in the age of chatbots. YouTube ads: \$11.1 bn; Google Cloud: \$24.8 bn. Google said in 2025 it handles over 5 trillion searches a year, about 1.25 trillion a quarter. So very roughly, $\$63\text{ bn} \div 1.25\text{ trillion} \approx$ **5 cents of revenue per search**, most of it from the small share of searches with buying intent.

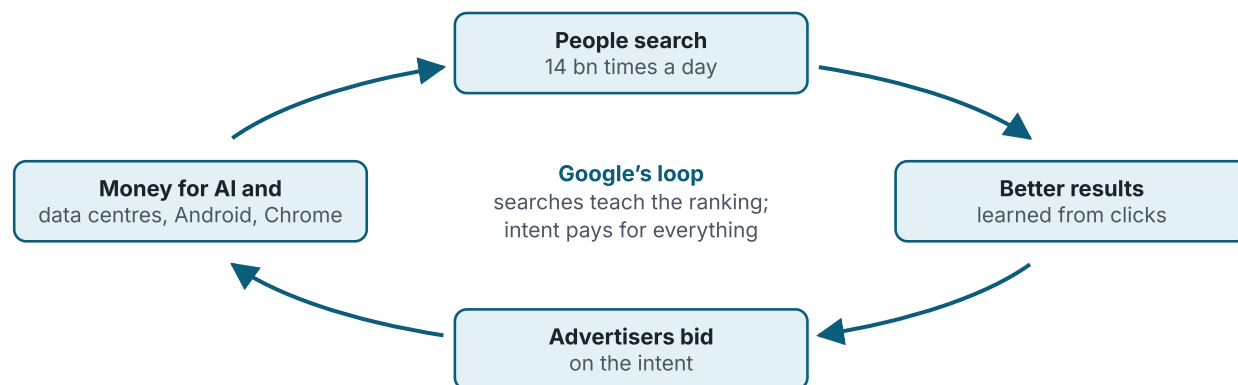


Figure 3.31 · Google's loop. Every search and click teaches the ranking what people find useful, which keeps people searching, which makes their intent worth more to advertisers. The ad money funds Android and Chrome, which bring in more searches. *Author's summary.*

Moat, in plain words. Being the default: the search box on Android, in Chrome, and (through payments to Apple) in Safari. Twenty-five years of click data that tell it which result satisfies which question. And advertisers who already run campaigns there. In August 2024 a US court found Google had illegally kept its search monopoly; in September 2025 it ordered remedies that stopped short of selling Chrome but barred exclusive default deals.

NORTH STAR

Searches that end with the user satisfied (no quick re-search or return to results).

INPUTS

Queries per user · AI Overview and AI Mode use · ad click-through rate · Lens searches.

GUARDRAILS

Ads per results page · clicks sent to publishers · results-page load time.

STRENGTHS

- ✓ Default search almost everywhere
- ✓ The highest-intent ad inventory in the world
- ✓ Its own AI models, chips and data centres

WEAKNESSES

- ✗ AI answers can cut clicks to the websites that feed it
- ✗ Chatbots compete for the “answer me” queries
- ✗ Antitrust limits on default deals

What I'd build next

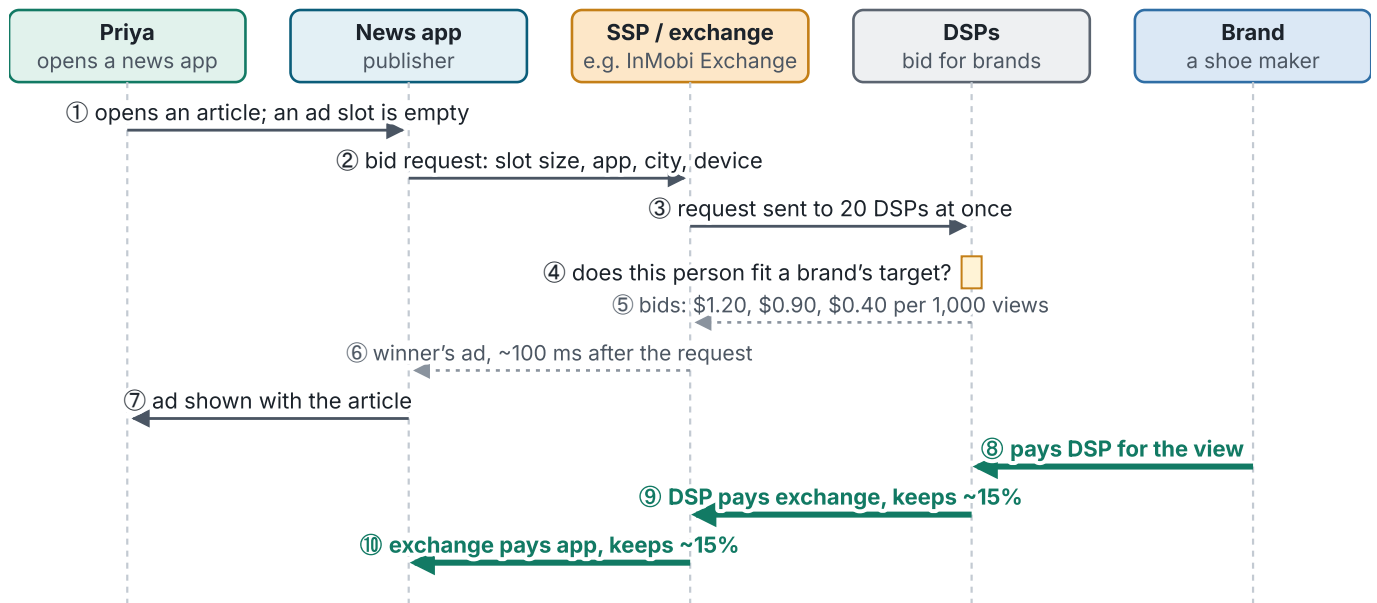
1. AI answers satisfy users but starve the websites that wrote the content → clearer source cards with a one-tap “read more” inside AI answers → clicks to publishers per AI answer, with satisfied searches held steady.
2. Small Indian shops can't write good ads → AI-made ads in Hindi and regional languages from a shop's Google Business Profile → new small advertisers; their spend after 90 days.

Ad tech: InMobi, Media.net, and the auction before the page loads

Walkthrough (illustrative persona). Priya, 29, opens a free news app on the metro. As the article loads, a banner for running shoes appears. She never sees what happened in the tenth of a second before: the app announced an empty ad slot, an exchange asked twenty bidders what they would pay to show Priya an ad, and a shoe brand's bidding software won at about \$1.20 per thousand views. The brand paid; three companies in between each kept a cut; the news app got the rest (Figure 3.32).

How the ad-tech stack works

Google, Meta and Amazon run **walled gardens**: they own the users, the ad slots and the buying tools, all in one company. Everyone else (news sites, games, small apps) sells ads through the **open web**, where specialist firms connect buyers and sellers. The advertiser uses a **DSP** (demand-side platform: software that bids for ads on its behalf). The app or site uses an **SSP** (supply-side platform: software that sells its ad slots to the highest bidder). An **ad exchange** is the marketplace in between. Each auction is **real-time bidding** (RTB): one auction per ad slot, finished in about 100 milliseconds.



Real-time bidding: an auction per ad slot, finished before the page has loaded. Middlemen keep a cut at every hop.

Figure 3.32 · One real-time auction. Steps ②–⑥ happen before the article finishes loading. The money in ⑧–⑩ flows back through every middleman, and each keeps a share. *Simplified from the OpenRTB protocol; bids and fee shares illustrative.*

One dollar of open-web ad spend, in cents (illustrative)

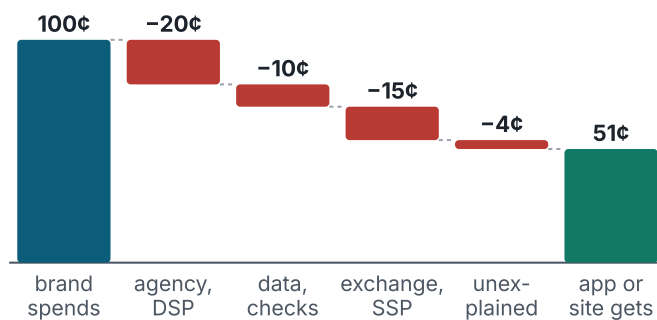


Figure 3.33 · Where a dollar of open-web ad spend goes. Roughly half reaches the app or site that showed the ad. *Illustrative split, consistent with the ISBA-PwC study of UK programmatic ads (2020), which traced about 51% to publishers.*

KEY IDEA · CONTEXTUAL VS PERSONAL TARGETING

Personal targeting shows ads based on who you are, built from tracking across apps and sites. **Contextual** targeting shows ads based on what you are reading: a marathon article gets running-shoe ads. Privacy changes (Apple's App Tracking Transparency in 2021; India's Digital Personal Data Protection Act, 2023) make personal tracking harder, so contextual ads, the old idea, are valuable again. Walled gardens suffer least: their users are logged in.

InMobi: India's first unicorn, now on the lock screen

Founded in Bengaluru in 2007, InMobi built a mobile ad network and became India's first startup valued at over \$1 bn (2011). It now runs two businesses. **InMobi Advertising** is a DSP and exchange for in-app ads, strongest in Asia.

Glance puts news, games and shopping cards on the lock screen of Android phones through deals with phone makers, so it reaches hundreds of millions of people who never downloaded it. That reach is its strength and its controversy: users didn't choose it. In 2025 the founders bought back SoftBank's stake to regain control, and in July 2026 InMobi appointed banks for an Indian listing of about \$1 bn. Its revenue is not public, and estimates vary widely.

Moat: deals with Android phone makers (a distribution channel competitors can't easily get) and in-app data across emerging markets. **Risk:** dependence on those deals, and competition from Google's and Meta's own app-install ads.

North Star: ad revenue per daily user reached (Glance) and spend through its exchange; **guardrails:** lock-screen opt-outs, ad fraud rate.

Media.net: ads that read the page, not the person

Founded in 2010 by Divyank Turakhia and run largely from Mumbai and Dubai, Media.net runs the contextual ad network behind Microsoft's and Yahoo's search ads on other websites. When you read an article about home loans on a news site, the "related searches" box and the ads beside it can come from Media.net: its software reads the page, works out its topic and picks matching ads. When a reader clicks, the advertiser's payment is shared between Media.net and the site. A Chinese consortium bought it in 2016 for about \$900 mn. **Moat:** page-understanding technology and a long partnership with Microsoft's ad network. **Risk:** that one partner is most of its business.

WHY OPEN-WEB AD TECH SURVIVES

✓ Millions of apps and sites need someone to sell their ads

WHY IT STRUGGLES

✗ Middlemen take about half of each dollar

- ✓ Brands want reach outside the walled gardens
- ✓ Privacy rules favour context over tracking

- ✗ Ad fraud: bots that “view” ads
- ✗ Google, Meta and Amazon take most new ad money

What I'd build next

1. Glance users feel it was forced on them → a first-unlock choice of topics (cricket, recipes, jobs) with one-tap off → 30-day retention of users who opted in; revenue per opted-in user.
2. Publishers don't trust where their ad money goes → a per-auction receipt showing each middleman's cut → publisher share of spend; publishers renewing.

Side by side

	Amazon India	Flipkart	Myntra	Google Search	InMobi · Media.net
For whom	Metro households, Prime members	Value buyers, Tier-2 India, phone upgraders	18–35, fashion and beauty	Everyone with a question	Apps and sites selling ad slots; brands buying reach
Earns from	Seller fees, fulfilment, ads, Prime	Seller and logistics fees, ads, fintech partners	Logistics fees, commission, ads	Ads on searches with intent	A cut of each ad auction or click
Latest number	Amazon ads \$19.8 bn (Q2 2026, worldwide)	FY25 revenue ₹20,493 Cr, loss ₹1,494 Cr	FY25 revenue ₹6,043 Cr, profit ₹548 Cr	Search \$63.3 bn (Q2 2026)	InMobi IPO of ~\$1 bn planned (2026)
Moat	Delivery network, Prime habit, purchase data	Phones, sale event, Ekart reach	Brand ties, fit data, young audience	Default search, click data	Phone-maker deals; page understanding
North Star	Weekly repeat purchases	Buyers active in 3+ months a year	Kept orders per customer	Satisfied searches	Revenue per user reached; spend through exchange
Biggest threat	Quick commerce taking frequent buys	Fighting Meesho and Blinkit at once	Returns; value fashion apps	AI chatbots; antitrust	Walled gardens; privacy rules

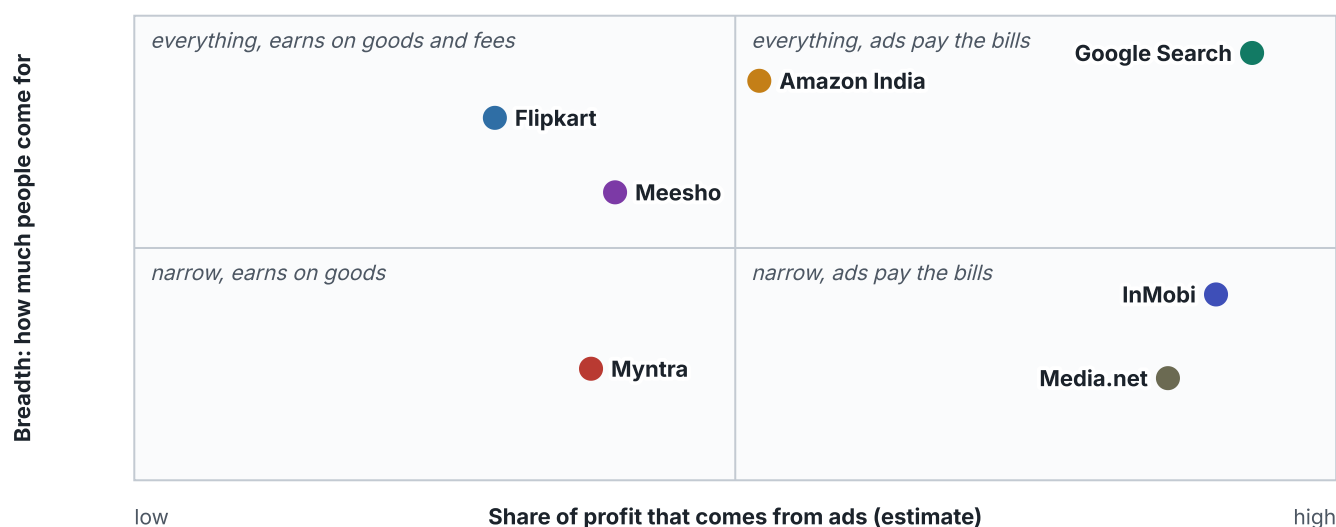


Figure 3.34 · Where the money comes from, and how much people come for. The shops sit on the left: they still earn mainly on goods and seller fees, with ads growing fast. Search and ad tech sit on the right: attention is what they sell. The shops are all drifting right, because ads are where their profit is. *Author's placement; positions are judgement, not measured shares.*

IN THE INTERVIEW · HOW THESE PRODUCTS CAME UP

- **“Why is Myntra profitable when Flipkart isn’t?”** One category, loyal young buyers who browse often, logistics fees from sellers and fast-growing ads; Flipkart funds a price war and a speed war at once.
- **“Should Amazon worry about quick commerce?”** Yes, for frequency: the everyday ₹200 orders that keep people opening the app. Not much for big, planned purchases. That is why Amazon Now exists.
- **“How does Google make money if AI answers the question?”** Most revenue comes from the small share of searches where people want to buy; AI answers mostly replace the informational ones. Watch clicks to advertisers and publishers.
- **“Explain real-time bidding to my grandmother.”** Every time a free app shows an ad, a tiny auction decides which advertiser pays most to show it to you, and it finishes before the page loads.

Machine learning and AI: what a PM needs to know

Open Myntra and almost everything you see was chosen by a model: the order of the home feed, the “similar products” row, the size hint, the search results, even which banner appears. None of these was written as rules by a person. Each was *learned* from millions of past clicks, purchases and returns. A PM doesn’t build these models, but decides where they are worth building, what data they need, what a wrong answer costs, and how to tell if they work.

AI, machine learning, deep learning and GenAI

These four words are nested, like Russian dolls (Figure 3.35). **Artificial intelligence** (AI) is any computer system doing something we would call intelligent, including hand-written rules (“if cart value > ₹499, delivery is free” is not AI; a chess program written as rules once was). **Machine learning** (ML) is AI that learns patterns from examples instead of being told the rules. **Deep learning** is ML using large neural networks, good at messy inputs like photos, speech and text. **Generative AI** (GenAI) is deep learning that *creates* new text, images or code, rather than choosing from existing options.

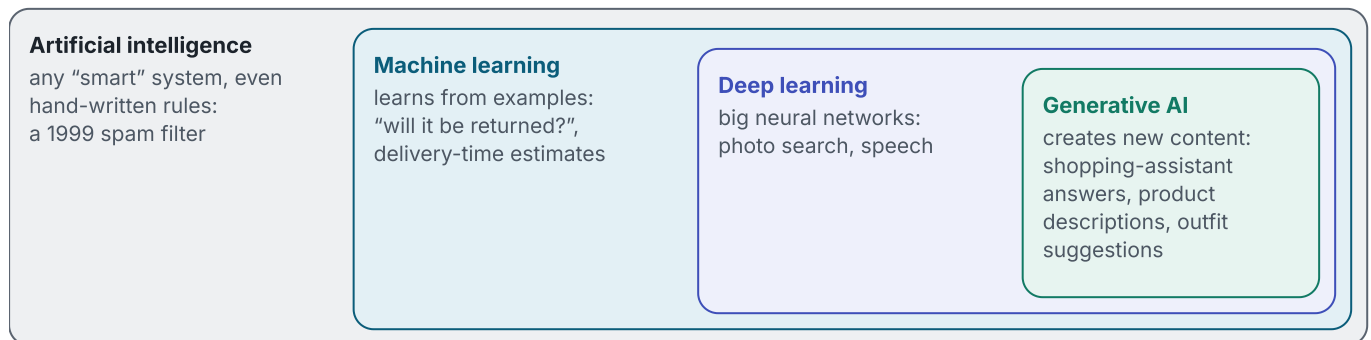


Figure 3.35 · Four nested ideas. Every GenAI feature is also deep learning, ML and AI, but most AI in an e-commerce app today is plain ML predicting a number: will she click, buy, return? *Author’s diagram; examples from Indian shopping apps.*

Three ways a machine learns

Supervised learning learns from examples where the right answer is known (**labelled** data). Show it 10 lakh past orders, each marked “returned” or “kept”, with details like size, brand, price and the customer’s history, and it learns to predict the chance that a *new* order will come back. **Unsupervised learning** has no right answers; it finds groups by itself. Give it every shopper’s behaviour and it may find five clusters, such as “sale-only phone buyers” or “weekly beauty browsers”, which marketing can then name and target. **Reinforcement learning** learns by trial and reward, like a child learning to cycle: it tries an action (which banner to show), sees the reward (a click or not), and slowly shifts towards what works (Figure 3.36).

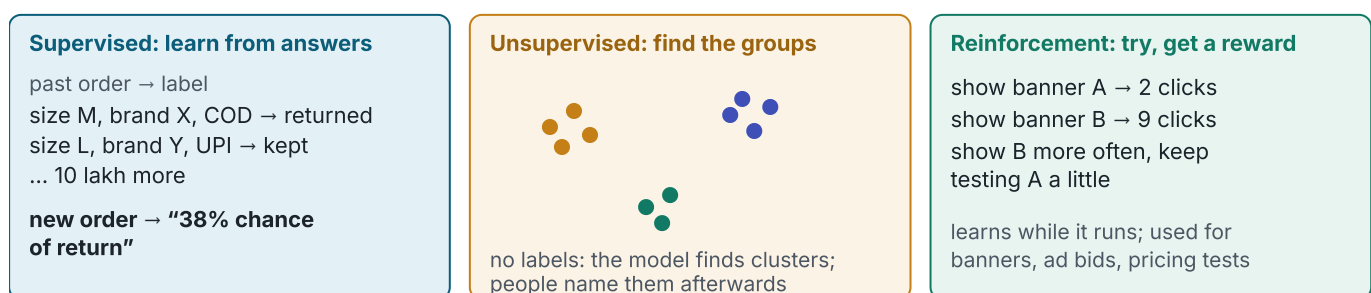
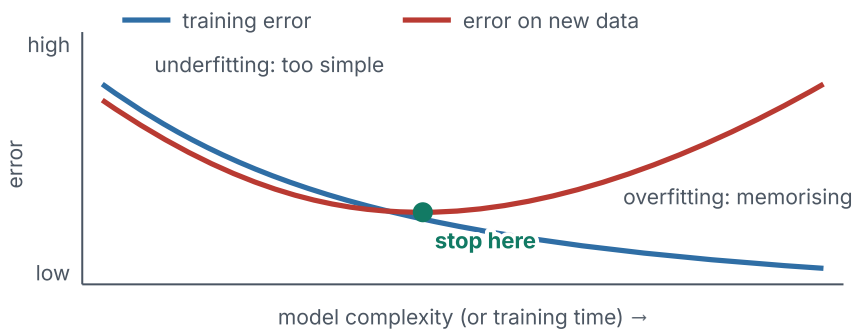


Figure 3.36 · Three ways a machine learns, on one shopping app. Most features a PM ships use supervised learning; it is the easiest to measure because the right answers are known. *Author’s examples; numbers illustrative.*

How a model is trained, and how it fools you

Training means showing the model labelled examples and letting it adjust itself to make fewer mistakes. The labelled data is split three ways: about 70% to **train** on, 15% for **validation** (to tune settings and choose between models) and 15% for a final **test** the model has never seen. The test score is the honest one.

Overfitting is the classic trap. A student who memorises last year’s question paper scores 100% on it and fails the real exam. A model can do the same: it learns quirks of the training data (a brand that happened to have many returns in one bad week) instead of the real pattern. Training error keeps falling, but error on new data starts rising (Figure 3.37). **Data leakage** is the sneakier trap: a column that secretly contains the answer. A return model that uses “refund issued” as an input looks 99% accurate in testing and is useless in real life, because at the moment of ordering nobody knows yet whether a refund will be issued.



Data leakage, a quick test

Ask of every input:
“would we know this at the
moment the prediction is
made?” “Refund issued”: no.
Remove it.

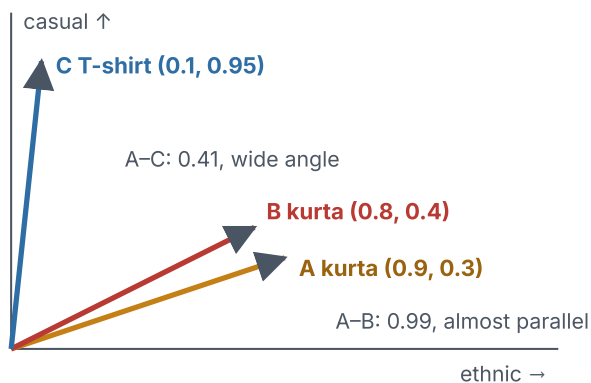
Figure 3.37 · When a model starts memorising. Error on the training data keeps falling as the model gets more complex, but error on new data turns up again. The best model sits at the bottom of the red curve, which is why a separate test set exists. *Standard illustration; shape only.*

Recommender systems, from the inside

The “you may also like” row is the most valuable model in e-commerce. It has one hard job: from a catalogue of, say, 50 lakh products, pick the 20 this person is most likely to want, in under 0.2 seconds. Two ideas sit underneath.

Collaborative filtering uses what similar people did: “people who bought this kurta also bought these juttis”. It needs no understanding of the product, only lots of behaviour. **Content-based filtering** uses the product’s own features: colour, fabric, style, price. It works for brand-new products that nobody has bought yet.

Embeddings. Modern systems turn every product and every user into a list of numbers, an **embedding**, so that similar things sit close together, like cities on a map. Similarity is then measured by **cosine similarity**: how closely two arrows point the same way (1 = same direction, 0 = unrelated). Worked example with two made-up dimensions, “ethnic” and “casual” (Figure 3.38): kurta A = (0.9, 0.3), kurta B = (0.8, 0.4), T-shirt C = (0.1, 0.95). $A \cdot B = 0.9 \times 0.8 + 0.3 \times 0.4 = 0.84$; divide by the lengths ($0.95 \times 0.89 \approx 0.85$): cosine ≈ 0.99 , almost the same direction. $A \cdot C = 0.09 + 0.29 = 0.38$; $\div (0.95 \times 0.96 \approx 0.91) \approx 0.41$. So B is recommended next to A, and C isn’t. Real embeddings have a few hundred dimensions, but the arithmetic is the same.



① Retrieve: 50 lakh → 500

nearest embeddings, fast, rough · ~20 ms

② Rank: 500 → 50

model predicts buy-and-keep · ~60 ms

③ Re-rank: 50 → 20

in stock, variety, ads · ~10 ms

20 products shown

whole trip must fit in ~150–200 ms

Figure 3.38 · Products as arrows. The smaller the angle between two arrows, the more similar the products, whatever their length. *Illustrative two-dimensional embeddings.*

Figure 3.39 · Three stages from 50 lakh products to one row. Cheap, rough filtering first; the expensive model only scores 500 items. *Common industry pattern; sizes and timings illustrative.*

The pipeline (Figure 3.39). Scoring 50 lakh products with a heavy model for every page view would take too long and cost too much. So systems work in stages: **retrieval** quickly pulls about 500 candidates whose embeddings are close to the user’s; **ranking** runs a detailed model on those 500 to predict the chance of a click, purchase or (better) a *kept* purchase; **re-ranking** applies business rules: hide out-of-stock items, avoid showing ten near-identical kurtas, place sponsored products in fixed slots. PMs own that last stage most directly.

Where it breaks	What happens	PM decision
Cold start	A new user or a new product has no history, so collaborative filtering has nothing to go on	Ask 3 quick preferences at sign-up; use product features for new items; give new items some guaranteed exposure
Popularity loop	Popular items get shown, so they get clicked, so they get shown more; small sellers vanish	Reserve slots for exploration; measure the share of sales going to new sellers
Wrong target	A model tuned for clicks recommends eye-catching items people return	Train on “bought and kept”, not on clicks
Offline vs online	A model that scores better on last month’s data can do worse with real users	Ship only after an A/B test on live traffic

Neural networks, in one picture

Visual search runs on a **neural network**: layers of simple units, each adding up its inputs with different weights and passing the result on. The first layers of an image model learn to spot edges and colours, the middle layers patterns such as embroidery or a collar, and the last layers whole ideas like “anarkali kurta” (Figure 3.40). **Training** means nudging millions of weights, a little at a time, so the network’s guesses on labelled photos get less wrong. Nobody writes the rule “an anarkali flares below the waist”; the network works it out from thousands of labelled photos.

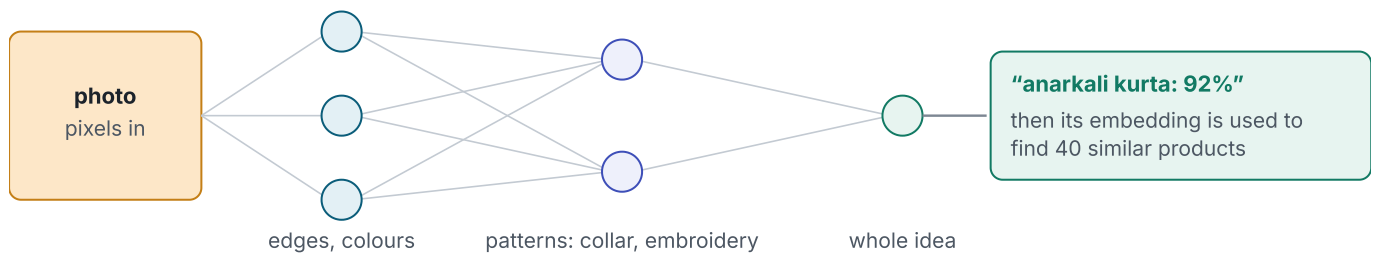


Figure 3.40 · A neural network looking at a photo. Each layer builds on the one before: from pixels to edges to patterns to a label. Real networks have dozens of layers and millions of connections. *Schematic.*

Generative AI: what changes for the PM

A classic model picks from options: a score, a label, a ranked list. A generative model **writes something new**: Amazon’s shopping assistant composes an answer from reviews; a seller tool writes a product description from a photo. Three things change. **Wrong answers look confident**: a generated answer can state a fabric that isn’t in the listing (a **hallucination**), so answers must be grounded in the product’s own data. **Every answer has a running cost**, paid per word generated, unlike a recommendation that is cheap once trained. And **quality is harder to measure**: there is no single right answer to compare against, so teams build test sets of questions with human ratings.

WHEN IT BREAKS

Models learn yesterday. After a big change (a new sale format, a pandemic, a price war) the patterns they learned stop holding, and accuracy falls silently; this is called **drift**. The only defence is monitoring live accuracy against fresh labels, and retraining on a schedule.

IN THE INTERVIEW

“I’d start with a simple rule or a popularity baseline, then use a model only if it beats that baseline in an A/B test by enough to cover its cost. I’d train it on the outcome we actually want, bought and kept, not on clicks.”

PM decision	What to ask the team
Build, buy or rule	Does a simple rule get 80% of the value? Is there an API (a ready-made model we can call) that is good enough?
Data	Do we have labels for the outcome we want? How long until we know the answer (a return is known only 2–3 weeks later)?
Errors	Which mistake costs more, a false alarm or a miss? (That sets the threshold, next section.)
Speed and cost	How many milliseconds can it take on the page? What does each prediction or answer cost at our traffic?
Proof	Which offline test, then which live A/B test, and what guardrail must not get worse?

Precision, recall and the threshold dial

Some customers abuse returns: they send back a used item, or a cheap copy instead of the original. A shopping app trains a model that gives every return request a **fraud score** from 0 to 1. Somebody has to decide the score above which a return is held for inspection instead of refunded instantly. That somebody is usually the PM, and the choice decides how many honest customers get annoyed and how many cheats get paid.

Four boxes: the confusion matrix

Take 10,000 return requests, of which 200 (2%) are really fraudulent. At a threshold of 0.35 the model flags 300. Every request lands in one of four boxes (Figure 3.41). **True positives**: fraud, correctly flagged (150). **False positives**: honest customers wrongly flagged (150). **False negatives**: fraud the model missed (50). **True negatives**: honest, correctly let through (9,650).

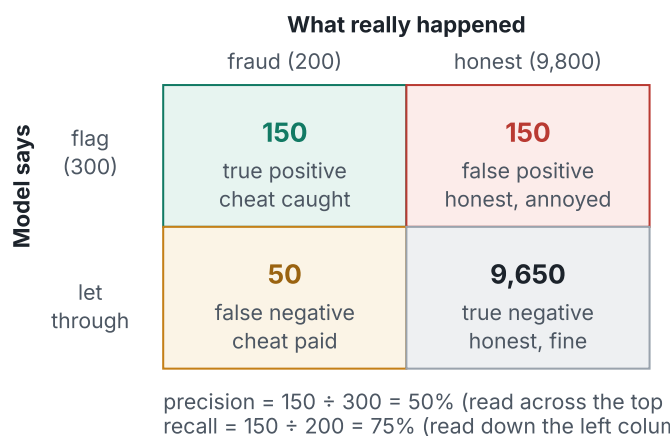


Figure 3.41 · The confusion matrix at threshold 0.35. Precision reads the top row; recall reads the left column. *Illustrative counts.*

The dial: one threshold, two kinds of mistake

Lower the threshold and the model flags more: it catches more cheats (recall up) but annoys more honest customers (precision down). Raise it and the opposite happens. There is no setting that is best at both (Figure 3.42). The right setting depends on **what each mistake costs**. Say a missed fraud costs ₹1,000 (the refund plus the lost item) and wrongly holding an honest customer's refund costs ₹300 (support calls, and some of them shop less afterwards).

Threshold	Flagged	Cheats caught	Precision	Recall	Missed × ₹1,000	False alarms × ₹300	Total cost
0.20 (strict)	600	185	31%	92%	₹15,000	₹1,24,500	₹1,39,500
0.35	300	150	50%	75%	₹50,000	₹45,000	₹95,000
0.50	180	120	67%	60%	₹80,000	₹18,000	₹98,000
0.65	120	100	83%	50%	₹1,00,000	₹6,000	₹1,06,000
0.80 (lenient)	60	55	92%	27%	₹1,45,000	₹1,500	₹1,46,500

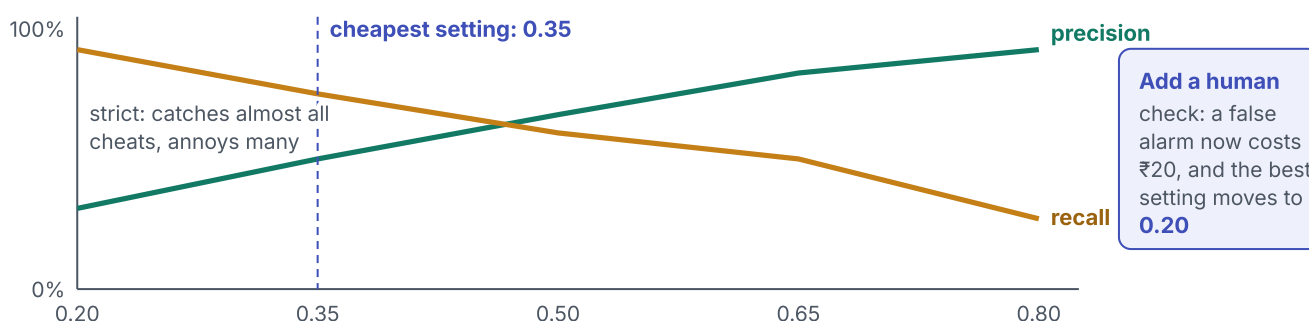


Figure 3.42 · The threshold dial. Precision rises and recall falls as the threshold goes up. The cheapest point depends on what each mistake costs, not on the model; change the costs (add a quick human review) and the best setting moves. *Illustrative model and costs; same numbers as the table.*

WHEN IT BREAKS

The labels arrive late: you learn a return was fraudulent only after the warehouse inspects it, weeks later, so today's precision is a guess. Cheats adapt: once a pattern is blocked they change it, and recall quietly drops. And one threshold for everyone can be unfair: a new customer in a small town with one odd return may be flagged far more often than a regular in a metro.

IN THE INTERVIEW

"Accuracy is misleading when fraud is 2% of cases. I'd look at precision and recall, put a rupee cost on each kind of mistake, and set the threshold where the total cost is lowest. For costly false alarms I'd add a human check instead of blocking outright."

PM decisions it drives. Which mistake is worse for this product (a spam filter hiding a real email, or a fraud filter letting a cheat through)? Whether flagged cases get blocked, slowed or sent to a person. Different thresholds for different groups, such as new vs long-standing customers, with a check that the difference is fair. And what to watch after launch: precision on reviewed cases, complaints from wrongly flagged customers, fraud losses.

The customer funnel and brand lift

Before the October sale, a fashion app spends ₹5 Cr on a video campaign on YouTube and Instagram (illustrative). A week later the CFO asks whether it worked. Sales are up, but they are always up before a sale, and most people

who saw the video won't buy for weeks. Clicks tell you little: few people click a brand video. The app needs a way to measure what the campaign did to people's *minds*, not just their fingers.

The funnel from "never heard of it" to "buys again"

Marketing thinks in a funnel of mental steps, each with its own measure (Figure 3.43). A campaign usually aims at one stage. A brand video aims at the top (people remember you and think well of you); a sale-day search ad aims at the bottom (people buy now). Judging a top-of-funnel campaign by next-day sales is like judging a school by one week's attendance: the effect is real but slow.

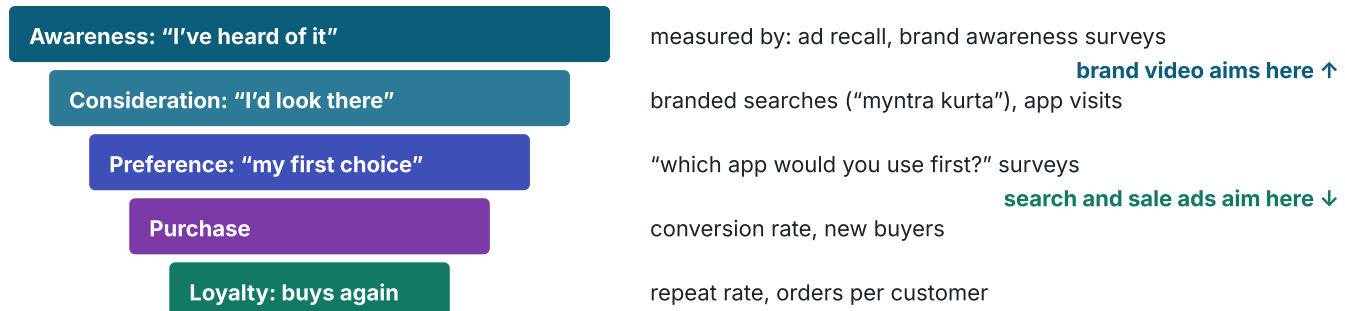
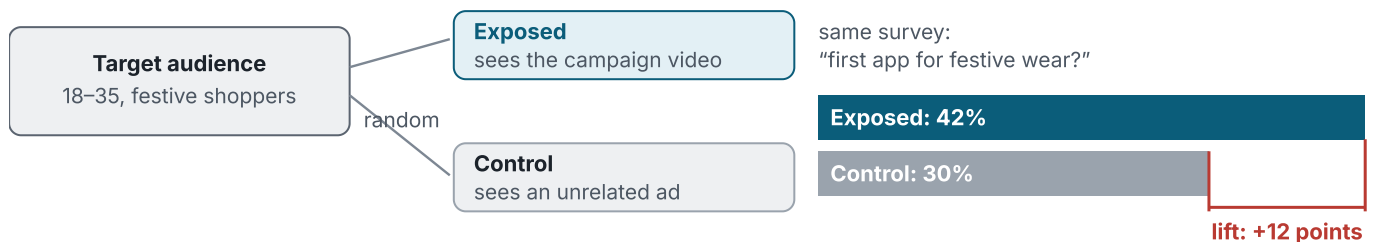


Figure 3.43 · The marketing funnel, and how each stage is measured. Upper stages are measured with surveys and searches, lower stages with sales. Match the measure to the stage the campaign targets. Adapted from IIM A Product Mindset 2026–27, pp. 121–122.

How a brand lift study works

A **brand lift study** measures the top of the funnel with a randomised experiment. Among the people the campaign targets, the ad platform (YouTube, Meta) randomly keeps some from seeing the ad: the **control** group sees whatever ad would have run otherwise. The rest, the **exposed** group, see the campaign. A few days later both groups get the same one-question survey inside the app: "Which of these fashion apps have you heard of?" or "Which would you use first for festive wear?". Because the groups were split at random, any difference comes from the campaign (Figure 3.44).

Worked example. 5,000 people in each group answer. In the control group 30% name the app when asked about festive wear; in the exposed group, 42%. **Absolute lift** = 42% – 30% = 12 points. **Relative lift** = 12 ÷ 30 = 40%. Is it real? With 5,000 answers per group the margin of error on the difference is about ±2 points, so a 12-point gap is far outside chance. Then the business sum: if the campaign reached 1 crore people, 12% of them, about 12 lakh extra people, now think of the app first, at about ₹40 each (₹5 Cr ÷ 12 lakh).



The only difference between the groups is the ad, so the 12-point gap is what the ad did (±2 points of chance, at 5,000 answers

Figure 3.44 · A brand lift study. Random assignment is what makes the comparison fair: the control group is the same kind of people, in the same week, minus the ad. Illustrative numbers; method as run by YouTube and Meta brand lift tools.

WHEN IT BREAKS

People say one thing and do another: a lift in "first choice" answers may never turn into orders. Survey takers are a special group (people who answer in-app polls). The control group can still see the brand elsewhere, on billboards or TV, which shrinks the measured lift. And a short survey can't capture effects that build over months.

IN THE INTERVIEW

"I'd first ask which stage of the funnel the campaign targets. For a brand video I'd run a brand lift study with a random control group and measure recall or preference; for a sale-day ad I'd measure incremental orders. I wouldn't judge a brand campaign by next-day clicks."

PM decisions it drives. Which funnel stage a feature or campaign is meant to move, and therefore which metric proves it. How big the control group must be (larger groups give a tighter margin of error but hide the ad from more people). Whether a lift is worth its cost per person persuaded. And a caution: a lift study proves the ad changed minds; only a sales test proves it changed wallets.

Connect the dots

Four ideas ran through the cases, the products and the technology. A PM at any shopping app meets all four in the same week, usually in the same meeting.

Thread	In the cases	In the products	In the technology
Only the kept order counts	Returns: split mix from rate, then fix size and fit; the outfit card's guardrail was returns on suggested items	Myntra's size hints; Amazon's next-day promise; Meesho's 36% gap between placed and delivered	Train recommenders on "bought and kept", not clicks; the fraud threshold set by the cost of each mistake
Ads pay the bills	Prime's value is partly the extra shopping members do	Myntra ₹915 Cr of ads; Amazon \$19.8 bn a quarter; Google's auction; open-web middlemen keep half	Re-ranking decides where sponsored items sit; brand lift proves what an ad did
Both sides must meet	Marketplace health is completed local matches, not messages	Zero commission to win sellers; Flipkart's sale assembles banks, sellers and refurbishers	Cold start and popularity loops decide which sellers ever get seen
Check before you build	"Does it need AI?"; break-even before a price cut; both guesstimates checked against a real number	Amazon's AI assistant grounded in reviews; Myntra's MyFashionGPT	Overfitting and leakage; offline scores confirmed by an A/B test

BACK TO JAIPUR · A SKETCH ANSWER FOR THE KURTA QUESTION

Is a 30% return rate on kurtas worth solving? Size it first. At about ₹250 lost per return and ₹195 earned per kept order, 10 kurtas at 30% returns earn ₹615; at 20%, ₹1,060. Ten points of return rate is worth about 70% more money per order, so yes, if the rate can move.

For whom? The platform (logistics cost), the seller (unsold stock that comes back creased) and the customer (a week of waiting and a trip to hand it back). All three gain, so it is a rare problem with no losing side.

How would you know? Split the rate: is it high because of the product (one brand's sizing), the customer (bracketing two sizes), or the season (wedding-wear mix)? Read the reason codes. Then act before the order, where a misfit costs ₹0: a fit hint from each product's returns, and size advice from the customer's own kept orders.

Goal (North Star): kept kurtas per 100 visitors. **Safety checks (guardrails):** conversion rate; complaints about refused returns. **Trade-off:** blocking returns lowers the rate and the sales together; only fixes that help people choose right lower one without the other.

KEY IDEA · THE E-COMMERCE PM'S THREE QUESTIONS

1. Does this change the number of orders that are *kept*, or only the number placed?
2. Who pays for it: the customer, the seller, a brand's ad budget or the platform?
3. Which side of the marketplace does it help, and does the other side lose?

IN THE INTERVIEW

In any shopping case, put a rupee number on one order within the first three minutes: "a ₹1,299 kurta leaves the app about ₹195 if kept and costs about ₹250 if returned". It shows you understand how the business earns before you change it.